

GG-GSW: Chosen-Ciphertext Secure Leveled FHE From Gadget Trapdoors

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Abstract. We build a leveled fully homomorphic encryption (FHE) scheme that achieves IND-CCA1 security under the learning with errors (LWE) assumption in the standard model. It is the first scheme of this kind that does not rely on succinct non-interactive arguments of knowledge (SNARK) to obtain security against active adversaries. Instead, we use the gadget lattice trapdoors introduced by Micciancio and Peikert [Eurocrypt 2012] in combination with a dual version of the GSW FHE scheme [Gentry, Sahai, Waters, Crypto 2013]. Instead of proving the integrity of a ciphertext with a SNARK, we use the gadget trapdoor to recover the LWE noise of a ciphertext. This ensures IND-CCA1 security because it allows us to determine whether a ciphertext queried to the decryption oracle will reveal information about the secret key to an adversary.

Our scheme is fully compact, multi-hop and requires very few changes to the original GSW scheme beyond the key generation and decryption algorithm. In particular, the homomorphic operations remain unchanged. We also follow ideas from Bourse et al. [Crypto 2016] to obtain IND-CPA^D security almost for free, without requiring correctness.

Keywords: Fully Homomorphic Encryption · Chosen-Ciphertext Security · Gadget Trapdoors

Important note: *The proof of Theorem 4.1 is incorrect, therefore the claims of IND-CCA1 security are currently unsupported. In short, game G_3 and G_4 are statistically distinguishable. We give more detail on the issue in Section 4.3. We thank an anonymous CRYPTO reviewer for pointing this out.*

The randomized evaluation algorithm from Section 5 may still be used with dual-GSW or GG-GSW to obtain IND-CPA^D security, but the proof of IND-CCA1^D security from Theorem 5.1 does not hold for the same reasons. A revised proof is in the works.

1 Introduction

Fully homomorphic encryption (FHE) is a powerful primitive that allows computation over encrypted data without needing access to the secret key. Since Gentry’s seminal paper [25], the field has seen dramatic improvements in the

efficiency and functionality of FHE schemes. See, e.g. [10,9,11,17,18,23] for some of the key developments. However, as FHE approaches usable performance for practical applications, it becomes increasingly important to consider its security in more advanced models.

Historically, the security of FHE schemes has overwhelmingly been analyzed in the context of passive adversaries, i.e. in the indistinguishability under chosen-plaintext attack (IND-CPA) model. While this is a natural starting point, this may be overly optimistic in practice. Among other things, the security of an IND-CPA-secure FHE scheme depends on the honesty of the party performing the homomorphic evaluation. Yet, this party can mount a host of practical attacks, ranging from simple distinguishing attacks to catastrophic key-recovery attacks. Furthermore, the difficulty of running such attacks is not necessarily correlated to their impact, e.g. a single leaked decryption of a ciphertext may allow an adversary to recover the secret-key in some cases [29].

Higher security models. Motivated by key-recovery attacks on approximate FHE schemes, Li and Micciancio [29] introduced the IND-CPA^D (IND-CPA with restricted decryptions) security notion. In addition to being able to observe encryptions and homomorphic evaluations as in the IND-CPA model, an adversary may query decryptions of honestly generated ciphertexts. While originally thought of as exclusively relevant for approximate FHE schemes, it was later shown that so-called “exact” FHE schemes suffer from similar attacks if their correctness guarantees are chosen too low [16,15].

Classically, the next target beyond IND-CPA security is indistinguishability under non-adaptive chosen-ciphertext attack (IND-CCA1). Here an adversary may see the decryption of arbitrarily chosen (possibly malformed) ciphertexts *before* receiving a distinguishing challenge. This is a well-studied notion, and security in this model already excludes key-recovery attacks against active adversaries.

Finally, beyond the CCA1 bound one may look at security against active adversaries with adaptive queries. It is well-known that the IND-CCA2 security model is not achievable for homomorphic encryption as it forbids any malleability. However, some intermediary models have been shown to be achievable. In this direction, Manulis and Nguyen [32] have introduced the vCCA notion, and recently Yang et al. [38] have introduced IV-CCA. Both models essentially allow an adversary to make adaptive decryption queries, but ensure that the ciphertexts do not depend on the challenge ciphertext.

Usual Techniques. Unfortunately, to date only a few techniques are known to provably achieve these higher levels of security, and each incur significant overhead.

A statistically correct CPA-secure FHE scheme achieves IND-CPA^D security [29]. While one may therefore consider that exact FHE schemes are beyond the scope of CPA^D style notions, we view achieving CPA^D security through correctness as its own method. Correctness is a delicate property for FHE schemes.

There is often a mismatch between theoretical upper bounds governing correctness guarantees and practical efficiency considerations. Using correctness to argue CPA^D security implies that it becomes a security property which must hold with overwhelming probability. However, in practice correctness is often seen as a performance property that can be tuned to achieve better efficiency. In particular, some applications may not require exponentially low decryption failure rates and a user may accept to set parameters such that decryption failures happen with noticeable probability e.g. 2^{-15} , to reduce the computational overhead. Additionally, libraries often use heuristic arguments to estimate correctness [3] in order to improve efficiency, which may be exploited by a malicious adversary [16].

Known CPA^D attacks rely on the dependence of the decryption algorithm on the LWE noise of a ciphertext. Li et al. [30] have shown that this can be mitigated through noise flooding at decryption. However this requires the noise flooding term to come from an exponentially wide distribution. Notice that the security of both methods, correctness and noise flooding, require correct upper bounds on the noise of a ciphertext, and their performance scales with the tightness of these bounds.

To obtain security against active adversaries i.e. CCA1 or the recently introduced vCCA or IV-CCA security, the main technique is to prove that ciphertexts are well-formed via proofs of knowledge [14,32,38]. The evaluator proves that the evaluation algorithm has been correctly executed on valid inputs. This significantly increases the computational overhead of evaluation, often relies on strong knowledge assumptions and also hurts the compactness of the underlying FHE schemes. The latest iteration [38] of this technique is based on the LWE assumption, but the proof still grows poly-logarithmically with the circuit size.

Outside of FHE. If we consider existing techniques to achieve security against active adversaries outside of the scope of FHE, the landscape is different. The well-known Fujisaki-Okamoto transform [24] is a highly efficient technique to obtain IND-CCA2 security, but is fundamentally incompatible with homomorphism by design. However, the (light) Cramer-Shoup encryption scheme, and the hash proof system framework that followed it [19,20] achieve CCA1 security by embedding an implicit proof of knowledge within the ciphertext. Most importantly for our purposes, it is an additively homomorphic encryption (AHE) scheme. Unfortunately, to the best of our knowledge no translation of the technique to lattice-based schemes exists, not even if we only ask for AHE schemes. In light of this, we ask the following questions.

Can we construct an FHE scheme that achieves IND-CCA1 security without relying on SNARKs? Can we achieve IND-CPA^D security without exponential noise flooding or statistical correctness?

1.1 Our Contributions

We answer both questions positively. Our main result is a leveled FHE scheme that achieves IND-CCA1 security under the LWE assumption in the standard model without requiring SNARKs. It is a variation of the (dual) GSW scheme [27], fully compact, multi-hop, and requires minimal changes to the existing constructions. In particular, ciphertext size scales very similarly to the original scheme and all homomorphic operations are unchanged. Along the way, we also provide the first CCA1-secure lattice-based AHE scheme that does not use proofs of knowledge to achieve security. It is a variation of the CCA1-secure encryption scheme given by Micciancio and Peikert [34].

Next, we extend the CPA^D model to the CCA1 setting. Recall that Brzuska et al. [13] have shown that vCCA security does not imply CPA^D security, hence CCA1 security also does not imply CPA^D security. Intuitively, an active adversary should always be stronger than a passive one, since it should have access to any information that a passive adversary does. Thus, as in [13], we introduce a CPA^D version of the IND-CCA1 game by augmenting the latter with the CPA^D oracles (cf. Figure 1).

Finally, we upgrade our CCA1 scheme to achieve IND-CCA1^D security over NC^1 , independently of its correctness guarantees. We obtain this “almost for free”, following techniques introduced by Bourse et al. in the context of circuit privacy [8]. We extend their randomized homomorphic evaluation algorithm for branching programs to the dual setting. We then prove the IND-CCA1^D security of the scheme for branching programs from the LWE assumption. As shown by Brakerski and Vaikuntanathan [12], this allows the use of GSW (and our scheme) to homomorphically evaluate any circuit in NC^1 .

Importantly, our security proof is fully independent of the correctness property of the scheme. In particular, it does not depend on the LWE modulus q . This allows us to reclaim some of the loss in the size of the parameters incurred to obtain IND-CCA1^D security, by trading off correctness for greater efficiency. Beyond this, it is possible to use our scheme with any (possibly heuristic) noise estimation strategy without compromising security. In particular, our scheme is robust against mistakes in the noise estimation. Any parameter set that yields a secure LWE problem may be used (although the scheme may not decrypt anymore).

1.2 Our Techniques

Our starting point is the CCA1-secure encryption scheme given by Micciancio and Peikert [34]. It is a dual-style encryption [26] that uses gadget trapdoors to control the noise. Simplifying away some details, a ciphertext has the form:

$$\mathbf{c} = \mathbf{s}^\top [\mathbf{A}_1 | \mathbf{U}\mathbf{G} - \mathbf{A}_1\mathbf{R}] + \mathbf{e}^\top + (\mathbf{0}, \mathbf{m}) \pmod{q}$$

where $\mathbf{R}$ is the secret key, $\mathbf{G} = (1, 2, \dots, 2^{\log q - 1})^\top \otimes \mathbf{I}_n \in \mathbb{Z}_q^{n \times n \log q}$ is the gadget matrix and $(\mathbf{s} \in \mathbb{Z}_q^n, \mathbf{U} \in \mathbb{Z}_q^{n \times n}, \mathbf{e} \in \mathbb{Z}_q^m)$ is the encryption randomness. Note that

$\mathbf{R}$ is chosen such that $\mathbf{A}_U = [\mathbf{A}_1 | \mathbf{U}\mathbf{G} - \mathbf{A}_1\mathbf{R}]$ is statistically close to uniformly random. Decryption happens by using the LWE inversion algorithm for gadget trapdoors from [34]. They prove CCA1 security by showing that decryptions are statistically independent of $\mathbf{R}$ and then use a puncturing argument. We refer the reader to [34] for more details.

This already nearly gives a CCA1-secure AHE scheme. However, sampling $\mathbf{U}$ randomly for every encryption breaks homomorphism. Our observation is that the only purpose of $\mathbf{U}$ is to be able to make the puncturing argument in the security reduction. We show that it is possible to prove the CCA1 security of the scheme under the LWE assumption through a careful analysis of the distribution of the decryption queries. Hence, we may fix $\mathbf{U} = \mathbf{I}_{n \times n}$ and obtain ciphertexts of the form:

$$\mathbf{c} = \mathbf{s}^\top [\mathbf{A}_1 | \mathbf{G} - \mathbf{A}_1\mathbf{R}] + \mathbf{e}^\top + (\mathbf{0}, \mathbf{m}) \pmod{q}.$$

We clearly obtain an AHE scheme, and we will show that we maintain CCA1 security. Taking this as a basis, we consider how to move to FHE. The structure of the gadget trapdoor is linear, so we cannot support BGV/BFV [12,9] style multiplications. However, the approximate eigenvector structure of the (dual) GSW scheme [27] appears as a natural candidate. Using the Alperin-Sheriff Peikert version [6], a dual GSW ciphertext is of the form:

$$\mathbf{C} = \mathbf{A}\mathbf{S} + \mathbf{E} + \mu\mathbf{G} \in \mathbb{Z}_q^{m \times m \log q}$$

where $\mathbf{G} = (1, 2, \dots, 2^{\log q - 1})^\top \otimes \mathbf{I}_m \in \mathbb{Z}_q^{m \times m \log q}$, $\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 \\ -\mathbf{R}\mathbf{A}_1 \end{bmatrix}$ is the public-key, and $\mathbf{R}$ is the secret-key. Now, if the matrix $\mathbf{A}$ had a gadget trapdoor similarly to our AHE scheme, we could prove the CCA1 security in the same way. This is exactly our approach; we embed a smaller (transposed) gadget matrix $\hat{\mathbf{G}} = (1, 2, \dots, 2^{\log q - 1}) \otimes \mathbf{I}_n \in \mathbb{Z}_q^{n \log q \times n}$ in the public-key $\mathbf{A}$. Consider now the new ciphertext:

$$\mathbf{C} = \begin{bmatrix} \mathbf{A}_1 \\ \hat{\mathbf{G}} - \mathbf{R}\mathbf{A}_1 \end{bmatrix} \mathbf{S} + \mathbf{E} + \mu\mathbf{G} \in \mathbb{Z}_q^{(m+n) \times (m+n) \log q}.$$

Decryption happens in two phases. Ignoring the details for now, the first phase recovers the message $\mu \in \mathbb{Z}_q$. In the second phase, we compute,

$$[\mathbf{R} | \mathbf{I}_{n \log q}](\mathbf{C} - \mu\mathbf{G}) = \hat{\mathbf{G}}\mathbf{S} + [\mathbf{R} | \mathbf{I}_{n \log q}]\mathbf{E}.$$

We then use the LWE gadget inversion algorithm to recover $\mathbf{E}$, and output the message μ if $\|\mathbf{E}\|$ is small enough. Clearly, the scheme is additive. For multiplication, we use the same operation as in the original scheme [6]. More precisely, for two ciphertexts $\mathbf{C}_1$ and $\mathbf{C}_2$ where $\mathbf{C}_i = \mathbf{A}\mathbf{S}^{(i)} + \mathbf{E}^{(i)} + \mu_i\mathbf{G}$, $\text{Mult}(\mathbf{C}_1, \mathbf{C}_2) = \mathbf{C}_1 \cdot \mathbf{X}$ where $\mathbf{X} \leftarrow \mathbf{G}^{-1}(\mathbf{C}_2)$ with $\mathbf{G}\mathbf{X} = \mathbf{C}_2$. Notice that our trapdoor $\mathbf{R}$ for the inversion algorithm is the same object as a (dual) GSW key. In particular, we observe the same approximate eigenvector phenomenon:

$$\begin{aligned}\mathbf{T}_R(\text{Mult}(\mathbf{C}_1, \mathbf{C}_2) - \mu_1\mu_2\mathbf{G}) &= \mathbf{T}_R(\mathbf{A}\mathbf{S}^{(1)} + \mathbf{E}^{(1)} + \mu_1\mathbf{G})\mathbf{X} - \mu_1\mu_2\mathbf{G} \\ &= \hat{\mathbf{G}}(\mathbf{S}_1\mathbf{X} + \mu_1\mathbf{S}_2) + \mathbf{T}_R(\mathbf{E}^{(1)}\mathbf{X} + \mu_1\mathbf{E}^{(2)}),\end{aligned}$$

where $\mathbf{T}_R = [\mathbf{R}|\mathbf{I}_{n \log q}]$. Hence, we obtain a (leveled) FHE scheme and can prove its CCA1 security in exactly the same way as the AHE construction. Since our scheme uses two gadget matrices $\mathbf{G}$, we call it GG-GSW for *double-gadget GSW*.

Proving IND-CCA1 security. We prove the IND-CCA1 security of the GG-GSW scheme using two main ideas. First, the distribution of the output of the decryption algorithm is independent of the secret key $\mathbf{R}$. We leverage this fact to introduce an inefficient decryption algorithm based on a statistical search-LWE oracle, and show that the distribution of its output is identical to that of the legitimate decryption algorithm. Second, since we only consider CCA1 security, the challenge ciphertext is only issued after access to the decryption oracle is revoked. In the reduction, this allows us to progressively modify the distribution of either the challenge ciphertext or the public key by alternating between using the legitimate decryption algorithm and the oracle-aided one.

Obtaining IND-CCA1^D security. To avoid basing security on correctness or using exponential noise-flooding, we construct a randomized evaluation algorithm for branching programs. This happens in two steps.

First, consider the randomized homomorphic evaluation algorithm introduced by Bourse et al. [8]. They take advantage of the particular noise growth properties of GSW over branching programs to construct a circuit-private scheme using only modest noise-flooding. In particular, they introduce the following rerandomization technique for GSW ciphertexts:

$$\mathbf{C}' = \mathbf{C}_{\text{gsw}} \cdot \mathbf{G}_{\text{rand}}^{-1}(\mathbf{V}) + \begin{pmatrix} \mathbf{0} \\ \mathbf{y}^\top \end{pmatrix}$$

where $\mathbf{G}_{\text{rand}}^{-1}(\mathbf{V})$ is chosen according to a discrete Gaussian with parameter r and $\mathbf{y}$ is a small noise flooding term. We will call this algorithm the BDMW randomization. They show that $\mathbf{C}'$ is a fresh encryption with noise distributed according to a discrete Gaussian with parameter $r\|\mathbf{e}\|$. Ignoring some details, they show that using this randomized multiplication procedure, the output of the homomorphic evaluation of a branching program is a fresh ciphertext with noise that only depends on the norms of the noise of the input ciphertexts.

We generalize their technique to the dual setting. This simply entails porting the relevant lemmas to support matrix noise $\mathbf{E}$ instead of vector noise. However, we cannot conclude IND-CCA1^D security yet, as the noise of the output still depends on the noise of the input ciphertexts in a noticeable way.

To remedy this, we use the HintLWE problem introduced by Kim et al. [28]. In this problem, an adversary must distinguish an LWE challenge given some noise hints, i.e. the adversary receives

$$(\mathbf{A}, \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q, \mathbf{e} + \tilde{\mathbf{y}})$$

and must distinguish $\mathbf{A}\mathbf{s} + \mathbf{e}$ from random. Kim et al. give a reduction between LWE and HintLWE given the correct distribution for $\tilde{\mathbf{y}}$. In our scheme, we will use this to our advantage to show that the dependence on $\mathbf{E}$ left by the BDMW randomization has no impact on the security of the input ciphertexts.

Putting it together, let us describe our randomized evaluation algorithm. On input some branching program Π and ciphertexts $\mathbf{C}_1, \dots, \mathbf{C}_l$, we first add the noise-flooding term $\mathbf{Y}_i$ to obtain $(\mathbf{C}'_1, \dots, \mathbf{C}'_l)$. We then evaluate the branching program using the BDMW procedure.

Ignoring some details, we are able to show that the noise of the evaluated ciphertext only depends on $\mathbf{E}_i + \mathbf{Y}_i$. From this, we prove IND-CCA1^D security using the HintLWE assumption.

1.3 Related Works

A recently published concurrent work by Ogilvie uses the HintLWE assumption to obtain IND-CPA^D and KR-CPA^D security [35]. Their work focuses on improving the analysis of noise-flooding techniques in the bit-security framework to reduce the overhead it generates. This is orthogonal to our work. They give an improved analysis for general LWE and RLWE schemes, which still requires exponential noise-flooding. On the other hand, we give a tailored approach to the GSW scheme and branching programs, which allows us to use polynomial noise-flooding. Recently, Libert [31] gave multiple proofs of CCA1 security for AHE schemes over small message spaces. They show that adding this constraint on the message space gives enough of an information-theoretical advantage to prove CCA1 security for schemes for which no proof is otherwise known such as Damgård's ElGamal [21].

2 Preliminaries

Notation. We write vectors in lower-case bold font (e.g. $\mathbf{x}$) and matrices in upper-case bold font (e.g. $\mathbf{A}$). For vectors $\mathbf{v}_1, \dots, \mathbf{v}_l$, we write $[\mathbf{v}_1 | \mathbf{v}_2 | \dots | \mathbf{v}_l]$ to denote the matrix whose columns are the $\mathbf{v}_i$. For matrices $\mathbf{A}_1, \dots, \mathbf{A}_l$, we write $\mathbf{A} = [\mathbf{A}_1; \dots; \mathbf{A}_l]$ to denote the matrix $\mathbf{A}$ that is the vertical concatenation of the $\mathbf{A}_i$, i.e. $\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 \\ \vdots \\ \mathbf{A}_l \end{bmatrix}$. For any set S , we write $x \xleftarrow{\$} S$ to denote that x is sampled from the uniform distribution over S . For any probability distribution $\mathcal{D}$, we write $x \leftarrow \mathcal{D}$ to denote that x is sampled according to the distribution $\mathcal{D}$.

2.1 Gaussian Distributions

For any $\sigma > 0$, the spherical Gaussian function on $\mathbb{R}^n$ with parameter σ is defined as $\rho_\sigma(\mathbf{x}) = \exp(-\pi \|\mathbf{x}\|^2 / \sigma^2)$ for all $\mathbf{x} \in \mathbb{R}^n$. For a rank- n matrix $\mathbf{S} \in \mathbb{R}^{m \times n}$ and

a vector $\mathbf{c} \in \mathbb{R}^n$, the ellipsoidal Gaussian function on $\mathbb{R}^n$ centered at $\mathbf{c}$ with parameter $\mathbf{S}$ is defined as:

$$\rho_{\mathbf{S},\mathbf{c}}(\mathbf{x}) = \exp(-\pi(\mathbf{x} - \mathbf{c})^\top (\mathbf{S}^\top \mathbf{S})^{-1} (\mathbf{x} - \mathbf{c}))$$

for all $\mathbf{x} \in \mathbb{R}^n$. Note that equivalently, $\rho_{\mathbf{S}}$ is the function with covariance matrix $\mathbf{\Sigma} = \mathbf{S}^\top \mathbf{S}$. The spherical case is obtained when $\mathbf{S} = \sigma \cdot \mathbf{I}_n$ and we write $\rho_{\sigma,\mathbf{c}}$. If $\mathbf{c} = \mathbf{0}$, we write $\rho_{\mathbf{S}}(\cdot)$. For a set A , $\rho_{\mathbf{S}}(A)$ denotes $\sum_{\mathbf{x} \in A} \rho_{\mathbf{S}}(\mathbf{x})$.

2.2 Lattices

An m -dimensional lattice Λ is a discrete additive subgroup of $\mathbb{R}^m$. For an integer $k \leq m$ and a rank k matrix $\mathbf{B} \in \mathbb{R}^{m \times k}$, $\Lambda(\mathbf{B}) = \{\mathbf{B}\mathbf{x} \in \mathbb{R}^m \mid \mathbf{x} \in \mathbb{Z}^k\}$ is the rank- k lattice generated by the columns of $\mathbf{B}$. We write Λ^* to denote the dual lattice $\{\mathbf{v} \in \mathbb{R}^n \mid \langle \mathbf{v}, \Lambda \rangle \in \mathbb{Z}\}$ and we write $\Lambda_q^\perp(\mathbf{B})$ to denote $\{\mathbf{v} \in \mathbb{Z}^m \mid \mathbf{B}\mathbf{v} = \mathbf{0} \bmod q\}$.

For a rank- n lattice Λ , $\mathbf{S} \in \mathbb{R}^{m \times n}$, and $\mathbf{c} \in \mathbb{R}^n$, the ellipsoidal Gaussian distribution with parameter $\mathbf{S}$ and support $\Lambda + \mathbf{c}$ is defined, for all $\mathbf{x} \in \Lambda + \mathbf{c}$, as:

$$\mathcal{D}_{\Lambda+\mathbf{c},\mathbf{S}}(\mathbf{x}) = \frac{\rho_{\mathbf{S}}(\mathbf{x})}{\rho_{\mathbf{S}}(\Lambda + \mathbf{c})}.$$

Definition 2.1 (Smoothing parameter [33]). For an n -dimensional lattice Λ and a positive real $\varepsilon > 0$, the smoothing parameter $\eta_\varepsilon(\Lambda)$ is the smallest s such that $\rho_{1/s}(\Lambda^* \setminus \{0\}) \leq \varepsilon$.

Lemma 2.1 ([33], Lemma 3.3). For any n -dimensional lattice Λ and $\varepsilon > 0$,

$$\eta_\varepsilon \leq \lambda_n(\Lambda) \sqrt{\frac{\ln(2n(1 + 1/\varepsilon))}{\pi}}$$

where $\lambda_n(\Lambda)$ is the smallest real number $r > 0$ such that $\dim(\text{span}(\Lambda \cap r\mathcal{B})) = n$ and $\mathcal{B}$ is the n -dimensional unit ball centered at the origin.

Lemma 2.2 ([34], Lemma 2.4). Let $n, m, q \geq 2$ with $q = p^e$ a power of a prime p . Let $\varepsilon > 0, \eta \geq \eta_\varepsilon(\mathbb{Z}^m)$ and $s > \eta$ be reals. Then for uniformly random $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, if

$$m \geq \max \left\{ n + \frac{\log(3 + 2/\varepsilon)}{\log p}, \frac{n \log q + \log(2 + 2/\varepsilon)}{\log(s/\eta)} \right\} \quad (1)$$

then $s \geq \eta_{2\varepsilon/\delta}(\Lambda_q^\perp(\mathbf{A}))$ except with probability at most δ . In particular, letting $s = 2\eta_\varepsilon(\mathbb{Z})$ and $\varepsilon = \text{negl}(n)$, $m = n \log q + \omega(\sqrt{\log n})$ fulfills Equation (1).

Lemma 2.3 ([7], Lemma 1.5). Let $\Lambda \subset \mathbb{R}^n$ be a lattice and $r \geq \eta_\varepsilon(\Lambda)$ for some $\varepsilon \in (0, 1)$. For any $\mathbf{c} \in \text{span } \Lambda$, we have

$$\Pr[\|\mathcal{D}_{\Lambda+\mathbf{c},r}\| \geq r\sqrt{n}] \leq 2^{-n} \cdot \frac{1+\varepsilon}{1-\varepsilon}.$$

If $\mathbf{c} = \mathbf{0}$ then the bounds holds for any $r > 0$ and $\varepsilon = 0$.

Lemma 2.4 (Simplified generalized leftover hash lemma [22]). *Let $\mathbf{e}$ be any random variable over $\mathbb{Z}_q^m$ and $f : \mathbb{Z}_q^m \rightarrow \mathbb{Z}_q^k$. Then,*

$$\Delta((\mathbf{X}\mathbf{e}, \mathbf{X}, f(\mathbf{e})), (r, \mathbf{X}, f(\mathbf{e}))) \leq \frac{1}{2} \sqrt{q^{n+k} \cdot 2^{-H_\infty(\mathbf{e})}}$$

where $\mathbf{X} \xleftarrow{\$} \mathbb{Z}_q^{n \times m}$ and $r \xleftarrow{\$} \mathbb{Z}_q^n$ and $H_\infty(\mathbf{e})$ is the min entropy of $\mathbf{e}$.

2.3 Matrices

For any matrix $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_m) \in \mathbb{R}^{n \times m}$, we write $\|\mathbf{B}\|$ to denote the maximum norm of the columns of $\mathbf{B}$. i.e. $\|\mathbf{B}\| = \max_i(\|\mathbf{b}_i\|)$. There exists a *singular value decomposition* $\mathbf{B} = \mathbf{Q}\mathbf{D}\mathbf{P}^\top$ with $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and $\mathbf{P} \in \mathbb{R}^{m \times m}$ are orthogonal matrices and $\mathbf{D} \in \mathbb{R}^{n \times m}$ is a rectangular diagonal matrix with nonnegative entries $s_i \geq 0$. The diagonal entries s_i are called the singular values of $\mathbf{B}$. The largest singular value of $\mathbf{B}$ is written $s_1(\mathbf{B})$ and the least singular value $s_n(\mathbf{B})$. We have $s_1(\mathbf{B}) = \max_{\mathbf{v}} \|\mathbf{B}\mathbf{v}\| \geq \|\mathbf{B}\|$ where the maximum is taken over the unit vectors $\mathbf{v} \in \mathbb{R}^m$.

Lemma 2.5 ([37], Theorem 4.4.5). *Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ be a subgaussian random matrix with parameter α . There exists a constant $C > 0$ such that for all $t \geq 0$ we have*

$$\Pr[s_1(\mathbf{A}) > C\alpha(\sqrt{m} + \sqrt{n} + t)] \leq 2\exp(-t^2)$$

Empirically, the constant C is a value close to $1/\sqrt{2\pi}$.

Lemma 2.6 ([2], Lemma 3). *For any $c \geq 1/\sqrt{2\pi}$, rank- n lattice Λ , $0 < \varepsilon < 1$, and matrix $\mathbf{S}$ such that $s_n(\mathbf{S}) \geq \eta_\varepsilon(\Lambda)$, we have*

$$\Pr[\|\mathcal{D}_{\Lambda, \mathbf{S}}\| \geq s_1(\mathbf{S})c\sqrt{n}] \leq \frac{1+\varepsilon}{1-\varepsilon} (c\sqrt{2\pi e} \cdot e^{-\pi c^2})^n.$$

2.4 The LWE Problem and Variations

We prove our schemes secure under the matrix version of the LWE assumption.

Definition 2.2 (LWE [36]). *Let n, m, q, k be positive integers. Let χ be a probability distribution over $\mathbb{Z}_q^{m \times k}$. The learning with error assumption, written $\text{LWE}_{n, m, q, \chi}^k$, asserts that for any PPT adversary $\mathcal{A}$, the following two distributions are computationally indistinguishable:*

$$(\mathbf{A}, \mathbf{A}\mathbf{S} + \mathbf{E} \bmod q) \approx_c (\mathbf{A}, \mathbf{U})$$

where $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$, $\mathbf{S} \xleftarrow{\$} \mathbb{Z}_q^{n \times k}$, $\mathbf{U} \xleftarrow{\$} \mathbb{Z}_q^{m \times k}$, $\mathbf{E} \leftarrow \chi$. More precisely for any PPT adversary $\mathcal{A}$, the following advantage must be negligible in the security parameter:

$$\text{Adv}_{\mathcal{A}}^{\text{LWE}}(\lambda) := |\Pr[\mathcal{A}(1^\lambda, \mathbf{A}, \mathbf{A}\mathbf{S} + \mathbf{E}) = 1] - \Pr[\mathcal{A}(1^\lambda, \mathbf{A}, \mathbf{U}) = 1]|$$

If the error distribution is the discrete spherical Gaussian with parameter α we may write $\text{LWE}_{n,m,q,\alpha}^k$.

Apart from the standard LWE problem, we will also use the HintLWE variant. In this setting, the adversary gets a tuple of hints to help with distinguishing the LWE challenge. This notion was introduced by Kim et al. [28] in the context of proofs of knowledge. They show that there is a reduction between the standard LWE problem and the HintLWE problem with slightly wider parameters. We give here a specialized version for our setting. We work in the plain (matrix) LWE setting instead of MLWE. Additionally, our hints only cover the noise term. Since the hints do not cover the LWE secret, we present the problem with uniform keys $\mathbf{S} \xleftarrow{\$} \mathbb{Z}_q^{n \times k}$, but any secret distribution that yields a secure LWE problem is possible.

Definition 2.3 (HintLWE). *Let n, m, q, k, t be positive integers. Let χ and ξ be probability distributions over $\mathbb{Z}_q^{m \times k}$ and $\mathcal{C}$ be a distribution over $\mathbb{Z}$. The HintLWE problem, denoted $\text{HintLWE}_{n,m,q,\chi}^{k,t,\xi,\mathcal{C}}$ is the task of distinguishing the following distributions:*

- $(\mathbf{A}, \mathbf{AS} + \mathbf{E} \bmod q, \gamma_1, \dots, \gamma_t, \mathbf{z}_1, \dots, \mathbf{z}_t)$ for $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$, $\mathbf{S} \xleftarrow{\$} \mathbb{Z}_q^{n \times k}$, $\mathbf{E} \leftarrow \chi$, $(\gamma_1, \dots, \gamma_t) \leftarrow \mathcal{C}^t$ and $\mathbf{z}_i = \gamma_i \cdot \mathbf{E} + \mathbf{Y}_i$ with $\mathbf{Y}_i \leftarrow \xi$ for $i \in [t]$.
- $(\mathbf{A}, \mathbf{B}, \gamma_1, \dots, \gamma_t, \mathbf{z}_1, \dots, \mathbf{z}_t)$ for $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$, $\mathbf{B} \xleftarrow{\$} \mathbb{Z}_q^{m \times k}$, $\mathbf{E} \leftarrow \chi$, $(\gamma_1, \dots, \gamma_t) \leftarrow \mathcal{C}^t$ and $\mathbf{z}_i = \gamma_i \cdot \mathbf{E} + \mathbf{Y}_i$ with $\mathbf{Y}_i \leftarrow \xi$ for $i \in [t]$.

The following theorem shows the hardness of matrix HintLWE under the matrix LWE assumption when the error is sampled from a discrete Gaussian distribution. This is just a direct adaptation of [28, Theorem 1] to the matrix setting.

Theorem 2.1. *Let n, m, q, k, t be positive integers and let $\mathcal{C}$ be a distribution over $\mathbb{Z}^t$. Let $B > 0$ be a real number which satisfies $\sum_{j=1}^t |\gamma_j|^2 \leq B$ for any possible $(\gamma_1, \dots, \gamma_t)$ sampled from $\mathcal{C}$. For $\sigma_1, \sigma_2 > 0$, let $\sigma > 0$ be a real number defined as $\frac{1}{\sigma^2} = 2(\frac{1}{\sigma_1^2} + \frac{B}{\sigma_2^2})$. If $\sigma \geq \sqrt{2} \cdot \eta_\varepsilon(\mathbb{Z})$ for $0 < \varepsilon \leq 1/2$, then there exists an efficient reduction from $\text{LWE}_{n,m,q,\sigma}^k$ to $\text{HintLWE}_{n,m,q,\sigma_1}^{k,t,\sigma_2,\mathcal{C}}$ that reduces the advantage by at most $2mk \cdot \varepsilon$.*

The proof is essentially unchanged from [28, Theorem 1]. In the original HintLWE problem statement, the roles of the secret and the noise are symmetric. We only need to generalize the statement to matrix noise. This follows from a simple vectorization argument and the rest of the reduction is unchanged.

2.5 Gadget Trapdoors

Let q be a positive integer and set $l = \lceil \log q \rceil$. Following [34], we call a vector $\mathbf{g}$ of the form $\mathbf{g} = (1, 2, \dots, 2^{l-1})$ a gadget vector. We similarly define the gadget matrix $\mathbf{G} = \mathbf{g}^\top \otimes \mathbf{I}_n = \text{diag}(\mathbf{g}^\top, \dots, \mathbf{g}^\top) \in \mathbb{Z}_q^{n \times nl}$.

We recall the gadget LWE inversion algorithm introduced by Micciancio and Peikert [34].

Lemma 2.7 (adapted from [34], Theorem 5.4). *Let n, k, m be positive integers and q be a power of two. Let $\hat{\mathbf{G}} = \mathbf{I}_n \otimes \mathbf{g} \in \mathbb{Z}_q^{n \cdot \log q \times n}$ be the (transposed) gadget matrix. Let $\mathbf{A}_1 \in \mathbb{Z}_q^{m \times n}$, $\mathbf{R} \in \mathbb{Z}^{(n \log q) \times m}$ and $\mathbf{T}_R = [\mathbf{R} | \mathbf{I}_{n \log q}]$. Let,*

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 \\ \hat{\mathbf{G}} - \mathbf{R}\mathbf{A}_1 \end{bmatrix}.$$

Let $\mathbf{B} = \mathbf{A}\mathbf{S} + \mathbf{E} \bmod q$ for any $\mathbf{S} \in \mathbb{Z}_q^{n \times k}$ and some $\mathbf{E} \in \mathbb{Z}_q^{(m+n \log q) \times k}$ such that $\|\mathbf{E}\| < q/4s'$ with $s' \leq \sqrt{s_1(\mathbf{R})^2 + 1}$. There exists a deterministic polynomial-time algorithm Invert that on input $(\mathbf{T}_R, \mathbf{A}, \mathbf{B})$, returns $(\mathbf{S}, \mathbf{E})$.

In short, the Invert algorithm first computes $[\mathbf{R} | \mathbf{I}_{n \log q}] \mathbf{A} = \hat{\mathbf{G}} \mathbf{S} + [\mathbf{R} | \mathbf{I}_{n \log q}] \mathbf{E}$. It then uses the particular structure of the gadget matrix to recover $(\mathbf{S}, \hat{\mathbf{E}})$, given that $[\mathbf{R} | \mathbf{I}_{n \log q}] \mathbf{E}$ is sufficiently short. Finally, it outputs $\mathbf{S}$ and $\mathbf{E} = \mathbf{B} - \mathbf{A}\mathbf{S}$. We refer the reader to [34] for the details. We abuse notation and call $\text{Invert}(\mathbf{T}_R, \mathbf{A}, \mathbf{B})$ both for matrices $\mathbf{A}$ of the form above, or the transposed version where $\mathbf{A}' = [\mathbf{A}'_1 | \mathbf{G} - \mathbf{A}'_1 \mathbf{R}']$.

The gadget matrix also allows to sample for any vector $\mathbf{u} \in \mathbb{Z}_q^n$, a short vector $\mathbf{x} \in \mathbb{Z}_q^m$ such that $\mathbf{G}\mathbf{x} = \mathbf{u}$.

Definition 2.4 (adapted from [34], [6], [8]). *There is a randomized, efficiently computable function $\mathbf{G}_{\text{rand}}^{-1}(\cdot) : \mathbb{Z}_q^n \rightarrow \mathbb{Z}_q^{n \lceil \log q \rceil}$ such that $\mathbf{x} \leftarrow \mathbf{G}_{\text{rand}}^{-1}(\mathbf{v})$ is drawn from a distribution close to a discrete Gaussian $\mathcal{D}_{\Lambda_q^\perp(\mathbf{G}) + \mathbf{G}_{\text{det}}^{-1}(\mathbf{v}), r}$, where $\mathbf{G}_{\text{det}}^{-1}(\cdot)$ denotes the deterministic bit decomposition. For any matrix $\mathbf{V} \in \mathbb{Z}_q^{n \times p}$, we similarly define $\mathbf{X} \leftarrow \mathbf{G}_{\text{rand}}^{-1}(\mathbf{V})$ to be the matrix in $\mathbb{Z}_q^{n \lceil \log q \rceil \times p}$ that has been obtained by applying $\mathbf{G}_{\text{rand}}^{-1}$ individually to the columns of $\mathbf{V}$. We write $\mathbf{G}^{-1}$ to denote that either $\mathbf{G}_{\text{det}}^{-1}$ or $\mathbf{G}_{\text{rand}}^{-1}$ can be used.*

2.6 Homomorphic Encryption

Definition 2.5 ((Fully) Homomorphic Encryption). *A public-key homomorphic encryption (HE) scheme $\text{HE} = (\text{Gen}, \text{Enc}, \text{Dec}, \text{Eval})$ for a family of functions $\mathcal{F}$ and a message space $\mathcal{M}$ is a quadruple of PPT algorithms such that:*

- $\text{Gen}(1^\lambda)$: Output a public key / secret key pair denoted (pk, sk) .
- $\text{Enc}_{\text{pk}}(m)$: On input a message $m \in \mathcal{M}$ and the public-key pk , output the encryption of m denoted $c \leftarrow \text{Enc}_{\text{pk}}(m)$. We write $c \leftarrow \text{Enc}_{\text{pk}}(m; r)$ to denote that c is an encryption of m with randomness r .
- $\text{Dec}_{\text{sk}}(c)$: On input a ciphertext c and the secret key sk , output $m \leftarrow \text{Dec}_{\text{sk}}(c)$ with $m \in \mathcal{M} \cup \{\perp\}$.
- $\text{Eval}_{\text{pk}}(f, c_1, \dots, c_l)$: The public key pk in the input is optional. On input a function $f \in \mathcal{F}$, and a tuple of ciphertexts $c_1, \dots, c_l$, output a ciphertext $\hat{c} \leftarrow \text{Eval}_{\text{pk}}(f, c_1, \dots, c_l)$.

HE is a *fully homomorphic encryption* (FHE) scheme if $\mathcal{F}$ is the set of all the efficiently computable functions. If for all functions $f \in \mathcal{F}$ and all plaintexts $m_1, \dots, m_l \in \mathcal{M}$, it holds that:

$$\Pr[\text{Dec}_{\text{sk}}(\text{Eval}_{\text{pk}}(f, c_1, \dots, c_l)) \neq f(m_1, \dots, m_l)] \leq \text{negl}(\lambda)$$

we say that HE is statistically correct. If the above always holds, we say that HE is perfectly correct. We depart from the convention of assuming that a so-called “exact” FHE scheme is statistically correct. We consider an exact FHE scheme to be a scheme where the expected result of a decryption is $f(m_1, \dots, m_l)$ as in the equation above. In particular, an exact scheme can have arbitrarily low correctness probability.

We use the standard definitions of IND-CPA and IND-CCA1. The definition of IND-CPA^D security may be found in [29]. We recall them in Appendix A.

3 Revisiting IND-CPA^D Security

As shown by Brzuska et al. [13], CCA1 security does not imply CPA^D security. The intuition behind this separation is that whenever the noise growth of an FHE scheme is message-dependent, an adversary may mount a distinguishing attack by submitting two challenge messages m_0 and m_1 that lead to different noise behaviors. This does not help in a CCA1 attack since the decryption oracle is disabled before the challenge bit is even drawn.

Lemma 3.1. *If there exists a statistically correct, IND-CCA1-secure HE scheme, then there exists an approximate HE scheme that is IND-CCA1 secure but not IND-CPA^D secure.*

Although this follows from [13], we provide a different proof in Appendix B since we will reuse the construction in the next section.

3.1 New Definitions

We define the IND-CCA1^D security game. Informally, the adversary gets access to the decryption oracle first. Once it is ready the game enters into a second phase, where the adversary now has access to the CPA^D encryption, decryption, and evaluation oracles.

Definition 3.1 (IND-CCA1^D). *Let $\Pi = (\text{Gen}, \text{Enc}, \text{Dec}, \text{Eval})$ be a public-key HE scheme. Consider the IND-CCA1^D experiment given in Figure 1. The advantage of an adversary $\mathcal{A}$ is given by:*

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{IND-CCA1}^D}(\lambda) = |\Pr[\text{Exp}_{\Pi, \mathcal{A}}^{\text{IND-CCA1}^D}(\lambda) = 1] - \frac{1}{2}|.$$

We say that Π is indistinguishable against non-adaptive chosen-ciphertext attack and passive decryption (IND-CCA1^D) if for all PPT adversaries $\mathcal{A}$, the advantage $\text{Adv}_{\Pi, \mathcal{A}}^{\text{IND-CCA1}^D}(\lambda)$ is negligible in λ .

$\text{Exp}_{\Pi, \mathcal{A}}^{\text{IND-CCA1}^D}(\lambda)$	$E_b(m_0, m_1)$ with $m_0, m_1 \in \mathcal{M}$
1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$	1 : $c \leftarrow \text{Enc}_{\text{pk}}(m_b)$
2 : $s \leftarrow \mathcal{A}^{\text{Dec}_{\text{sk}}(\cdot)}(\text{pk})$	2 : $S \leftarrow S \cup (m_0, m_1, c)$
3 : $b \xleftarrow{\$} \{0, 1\}, S := \emptyset$	3 : return c
4 : $b' \leftarrow \mathcal{A}^{E_b(\cdot), H(\cdot), D(\cdot)}(s)$	
5 : return $(b \stackrel{?}{=} b')$	
$\text{Dec}_{\text{sk}}(c)$	$D(j)$ with $j \in \{1, \dots, S \}$
1 : return $\text{Dec}_{\text{sk}}(c)$	1 : if $S[j].m_0 \neq S[j].m_1$
	2 : then return $\perp$
	3 : else
	4 : return $\text{Dec}_{\text{sk}}(S[j].c)$
$H(f, J)$ with $f \in \mathcal{F}$ and $J = (j_1, \dots, j_l)$	
1 : $\hat{c} \leftarrow \text{Eval}(f, S[j_1].c, \dots, S[j_l].c)$	
2 : $S \leftarrow S \cup (f(S[j_1].m_0, \dots, S[j_l].m_0), f(S[j_1].m_1, \dots, S[j_l].m_1), \hat{c})$	
3 : return $\hat{c}$	

Fig. 1. The IND-CCA1^D experiment $\text{Exp}_{\Pi, \mathcal{A}}^{\text{IND-CCA1}^D}(\lambda)$, with the unrestricted decryption oracle $\text{Dec}_{\text{sk}}(\cdot)$ and the CPA^D oracles $E_b(m_0, m_1)$, $H(f, J)$, and $D(j)$.

The following lemma shows that IND-CCA1 and IND-CPA^D security do not imply IND-CCA1^D security. Indeed, the CCA1 oracle may uncover a partial hint to an attacker that requires the CPA^D decryption oracle to complete.

Lemma 3.2. *If there exists a statistically correct, IND-CCA1 secure HE scheme, then there exists an HE scheme that is both IND-CCA1 secure and IND-CPA^D secure, but not IND-CCA1^D secure.*

Proof. We use a similar construction to the one used in the proof of Lemma 3.1. However, we now puncture the decryption algorithm on a single message m_w . We place an encryption c_w of m_w as a hint in the public key. Decrypting this hint requires the unrestricted decryption algorithm, as an IND-CPA^D adversary cannot get c_w to be in the state of the restricted decryption oracle. On the other hand, using this hint is only possible in the challenge phase, as the attack exploits the message-dependent noise to distinguish. In more detail:

Let Π' be as in the proof of Lemma 3.1. Without loss of generality, suppose that the size of message space $\mathcal{M}$ is exponential in the security parameter. If not, we can simply provide the hint in the form of encryptions of the bit decomposition of m_w , adapting the encryption, evaluation, and decryption algorithm accordingly.

We construct Π in a similar manner to Lemma 3.1, but change the key generation algorithm and the decryption algorithm as follows:

- $\text{Gen}(1^\lambda)$: Sample a key pair $(\text{pk}', \text{sk}') \leftarrow \text{Gen}'(1^\lambda)$. Sample a random message $m_w \xleftarrow{\$} \mathcal{M}$ and compute $c_w \leftarrow \text{Enc}'_{\text{pk}'}(m_w)$. Return the public-key $\text{pk} = (\text{pk}', c_w)$ and the secret-key $\text{sk} = (\text{sk}', m_w)$.
- $\text{Dec}(c)$: Parse c as (c_0, c_1) . Compute $m = \text{Dec}(c_0)$ and $d = \text{Dec}(c_1)$. If either m or d is the $\perp$ symbol, return $\perp$. If $d = m_w$ return $m + \alpha$, where $\alpha \neq 0$ is a fixed public element. Else return m .

Again, the scheme remains IND-CCA1 secure because α is independent of the secret key and public. The reduction chooses m_w itself, so it has no issues answering all decryption queries. Furthermore, from the point of view of a IND-CPA^D adversary, the scheme is perfectly correct. An adversary cannot get an arbitrary ciphertext in its state, as this would contradict IND-CPA security. In particular, it cannot have c_w , or any homomorphic evaluation of it as part of its state. Recall that for perfectly correct schemes, IND-CPA^D security is equivalent to CPA security.

However, in the IND-CCA1^D game an adversary can query its unrestricted decryption oracle with the ciphertext $c = (c_w, c')$, where $c' \leftarrow \text{Enc}_{\text{pk}'}(m')$ is an encryption of any arbitrary message $m' \neq m_w$ to obtain m_w . The rest of the attack proceeds as in the proof of Lemma 3.1, with $B = m_w$. $\square$

Lemma 3.3. *Any public-key FHE scheme that is IND-CCA1 secure and statistically correct achieves IND-CCA1^D security.*

Proof. The proof is very close to the one proving that IND-CPA security and statistical correctness implies IND-CPA^D security [29]. The distinction is that the reduction uses its CCA1 oracle to answer the decryption queries of the first phase. The second phase then runs the reduction from the CPA^D case. $\square$

4 CCA1 Secure FHE Scheme From Gadget Trapdoors

We begin by introducing a CCA1-secure AHE scheme based on the CCA1-secure encryption scheme by Micciancio and Peikert [34]. We then present our main construction, a leveled FHE scheme that combines the AHE scheme with the GSW message encoding [27,6].

4.1 A CCA1 Secure AHE Scheme

Let n, m, q be integers, and α a Gaussian parameter that define an LWE problem. Let s be another Gaussian parameter, p a plaintext modulus and $D > 0$ some bound. We leave the details of the choices of these values to the presentation of our main construction.

Our AHE scheme is the CCA1-secure encryption scheme from [34] with two modifications. First, we use a standard message encoding for lattice-based encryption, which allows to simplify the scheme and remove the mod $2q$ arithmetic.

Second, we remove the tag $\mathbf{U}$ that was used as a puncturing tool in the original scheme.

- **AHE.Gen**(1^λ): Choose $\mathbf{A}_1 \leftarrow \mathbb{Z}_q^{n \times m}$, $\mathbf{R} \leftarrow \mathcal{D}_{\mathbb{Z},s}^{m \times n \lceil \log q \rceil}$ and $\mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z},s}^m$. Compute $\mathbf{A}_2 = \mathbf{G} - \mathbf{A}_1 \mathbf{R} \pmod{q}$, $\mathbf{a}_3 = -\mathbf{A}_1 \mathbf{x} \pmod{q}$. Return the public key/secret-key pair,

$$\text{pk} := \mathbf{A} = [\mathbf{A}_1 | \mathbf{A}_2 | \mathbf{a}_3], \quad \text{sk} := (\mathbf{R}, \mathbf{x}).$$

- **AHE.Enc**($\text{pk}, \mu \in \mathbb{Z}_p$): Sample $\mathbf{z} \xleftarrow{\$} \mathbb{Z}_q^n$, $\mathbf{e}_1 \leftarrow \mathcal{D}_{\mathbb{Z},\alpha}^m$, $\mathbf{e}_2 \leftarrow \mathcal{D}_{\mathbb{Z},\alpha'}^{n \lceil \log q \rceil}$, $e_3 \leftarrow \mathcal{D}_{\mathbb{Z},\alpha}$ where $\alpha'^2 = \|e_1\|^2 + \alpha^2 m$ and set $\mathbf{e} = (\mathbf{e}_1, \mathbf{e}_2, e_3)$. Return

$$\mathbf{c}^\top = \mathbf{z}^\top \mathbf{A} + \mathbf{e}^\top + (\mathbf{0}, \Delta \mu)^\top \pmod{q}$$

where $\mathbf{0}$ has dimension $m + n \lceil \log q \rceil$ and $\Delta = \frac{q}{p}$.

- **AHE.Dec**($\text{sk} = (\mathbf{R}, \mathbf{x})$, $\mathbf{c} = (\mathbf{c}_1, \mathbf{c}_2, c_3) \in \mathbb{Z}_q^m \times \mathbb{Z}_q^{n \lceil \log q \rceil} \times \mathbb{Z}_q$):
 1. Call **Invert**($\mathbf{T}_\mathbf{R}, \mathbf{A}, (\mathbf{c}_1, \mathbf{c}_2)$) where $\mathbf{T}_\mathbf{R} = [\mathbf{R}; \mathbf{I}_m]$ to recover values $\mathbf{z}, \mathbf{e} = (\mathbf{e}_1, \mathbf{e}_2)$ such that $(\mathbf{c}_1, \mathbf{c}_2)^\top = \mathbf{z}^\top \mathbf{A} + \mathbf{e}^\top \pmod{q}$.
 2. If $\|\mathbf{e}\| \geq D$, return $\perp$.
 3. Return $\lfloor \Delta^{-1}(\mathbf{c}_3 - \langle \mathbf{c}_1 - \mathbf{e}_1, \mathbf{x} \rangle) \rfloor$.

Proposition 4.1. *The scheme above is correct, additively homomorphic and achieves IND-CCA1 security under the LWE assumption.*

We postpone the proof to the next section, as both the AHE scheme and the main FHE construction have a similar structure.

4.2 CCA1-Secure FHE Scheme

Our CCA1-secure leveled FHE scheme combines the key generation and decryption ideas from our AHE scheme with the gadget-based message encoding and homomorphic multiplication from the GSW scheme.

Let $n > 0$ and q be a power of 2, we set $l = \log q$. Let $\alpha > 0$ be a Gaussian parameter. Let $m = nl$ and set $\tilde{m} = 2m + 1$ and $k = \tilde{m} \cdot l$. The secret key generation uses a Gaussian parameter $s = \Omega(m\alpha\lambda\sqrt{k})$. The scheme supports a message space $\mathcal{M} \subset \mathbb{Z}_q$, but GSW is most often used with a binary message space $\{0, 1\}$ as this gives better noise growth properties. Therefore we present the binary message space version and point out that it generalizes to $\mathcal{M} \subset \mathbb{Z}_q$ through standard techniques [27].

Let $\mathbf{g} = (1, 2, \dots, 2^{l-1})$ be the gadget vector. Our scheme uses a gadget matrix $\mathbf{G} = \mathbf{I}_{\tilde{m}} \otimes \mathbf{g}^\top \in \mathbb{Z}_q^{\tilde{m} \times k}$ for message encoding and a transposed gadget matrix (of different dimension) $\hat{\mathbf{G}} = \mathbf{I}_n \otimes \mathbf{g} \in \mathbb{Z}_q^{m \times n}$ for the gadget trapdoor to ensure CCA1 security.

Let $D < \min \left\{ \frac{q}{8\sqrt{s^2 \cdot m + 1}}, \frac{q}{4B} \right\}$ with $B = C \cdot s \cdot (2\sqrt{m} + \sqrt{\lambda}) + 1$ where C is the constant from Lemma 2.5. Note that for any relevant parameter set, the left bound is the minimum. However it will help during the security proof to have both bounds.

Our CCA1 leveled FHE scheme GG-GSW is defined as follows:

- GG-GSW.Gen(1^λ): Sample $\mathbf{A}_1 \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$, $\mathbf{R} \leftarrow \mathcal{D}_{\mathbb{Z}^{m \times m}, s}$ and $\mathbf{x} \xleftarrow{\$} \mathcal{D}_{\mathbb{Z}^m, s}$. Compute $\mathbf{A}_2 = \hat{\mathbf{G}} - \mathbf{R}\mathbf{A}_1 \bmod q$ and $\mathbf{a}_3^\top = -\mathbf{x}^\top \mathbf{A}_1 \bmod q$. Return the public key/secret key pair,

$$\text{pk} = \mathbf{A} := \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \\ \mathbf{a}_3^\top \end{bmatrix} \in \mathbb{Z}_q^{\tilde{m} \times n} \quad \text{sk} := (\mathbf{R}, \mathbf{x})$$

- GG-GSW.Enc(pk = $\mathbf{A}$, $\mu \in \{0, 1\}$): Sample $\mathbf{S} \xleftarrow{\$} \mathbb{Z}_q^{n \times k}$, $\mathbf{E}_1 \leftarrow \mathcal{D}_{\mathbb{Z}^m, \alpha}^k$, $\mathbf{E}_2 \leftarrow \mathcal{D}_{\mathbb{Z}^m, s \hat{\mathbf{E}}^\top}^k$, $\mathbf{e}_3 \leftarrow \mathcal{D}_{\mathbb{Z}, \alpha}^k$, where $\hat{\mathbf{E}} = [\mathbf{E}_1 | \mathbf{I}_m]$. Set $\mathbf{E} = [\mathbf{E}_1; \mathbf{E}_2; \mathbf{e}_3^\top] \in \mathbb{Z}^{\tilde{m} \times k}$, the vertical concatenation of the noise terms. Return,

$$\mathbf{C} = \mathbf{AS} + \mathbf{E} + \mu \mathbf{G} \bmod q$$

- GG-GSW.Dec(sk, $\mathbf{C}$):
 1. Set $\mathbf{t}_\mathbf{x} = (\mathbf{x}, \mathbf{0}^m, 1) \in \mathbb{Z}_q^{\tilde{m}}$ and $\mathbf{T}_\mathbf{R} = [\mathbf{R} | \mathbf{I}_m | \mathbf{0}^m] \in \mathbb{Z}_q^{m \times \tilde{m}}$.
 2. Compute $\mathbf{y} = \mathbf{t}_\mathbf{x}^\top \mathbf{C} \bmod q$. Then set $\mu = \lfloor \frac{4}{q} y_i \rfloor$ where y_i is the penultimate coefficient of $\mathbf{y}$.
 3. Call Invert($\mathbf{T}_\mathbf{R}, \mathbf{A}, \mathbf{C} - \mu \mathbf{G}$) to recover values $\mathbf{S}, \mathbf{E}$ such that $\mathbf{C} = \mathbf{AS} + \mathbf{E} + \mu \mathbf{G} \bmod q$. If this step fails, return $\perp$.
 4. If $\|\mathbf{E}\| \geq D$ return $\perp$.
 5. Return μ .

We are using uniform LWE secrets $\mathbf{S} \xleftarrow{\$} \mathbb{Z}_q^{n \times k}$. As we will see, this does not impact the noise growth properties of the scheme and therefore this choice has no significant negative effect on its efficiency. However it will be useful in the security proof of our IND-CCA1^D construction. For the CCA1 scheme, one may choose any key distribution that yields a hard LWE problem.

Lemma 4.1 (Correctness). *Let $\mathbf{C} = \mathbf{AS} + \mathbf{E} + \mu \mathbf{G} \bmod q$ for any $\mathbf{S} \in \mathbb{Z}_q^{n \times k}$ and $\mu \in \{0, 1\}$. Then the decryption algorithm returns μ if $\|\mathbf{E}\| < D$.*

In particular, let $q = \Omega(s^2 \alpha m \sqrt{k} \cdot \log n)$. If $(\text{pk}, \text{sk}) \leftarrow \text{GG-GSW.Gen}(1^\lambda)$ and $\mathbf{C} \leftarrow \text{GG-GSW.Enc}(\text{pk}, \mu)$, we have $\mu \leftarrow \text{GG-GSW.Dec}(\text{sk}, \mathbf{C})$ with overwhelming probability over the randomness of the key generation and encryption algorithm.

Proof. The correctness of the scheme follows from combining the arguments of the correctness of the (dual) GSW scheme, that ensures the success of Step 2, and the correctness of the Invert procedure, which ensures the success of Step 3 and 4. More precisely, let $(\text{pk} = \mathbf{A}, \text{sk} = (\mathbf{R}, \mathbf{x})) \leftarrow \text{GG-GSW.Gen}(1^\lambda)$. We have that,

$$\|\mathbf{x}\| \leq s\sqrt{m} \quad \text{and} \quad s_1(\mathbf{R}) \leq Cs \left(2\sqrt{m} + \sqrt{\lambda} \right) \quad (2)$$

except with probability $2^{-\Omega(\lambda)}$, where the first inequality follows from Lemma 2.3 and the second comes from Lemma 2.5.

Let $\mathbf{C} = \mathbf{AS} + \mathbf{E} + \mu \mathbf{G} \bmod q$ be as in the statement of the lemma. We have $\mathbf{y} = \mathbf{t}_\mathbf{x}^\top \mathbf{C} = \mathbf{t}_\mathbf{x}^\top \mathbf{E} + \mu \mathbf{t}_\mathbf{x}^\top \mathbf{G}$. Hence Step 2 is correct if the magnitude of $\langle \mathbf{t}_\mathbf{x}, \mathbf{e} \rangle$

is less than $q/8$, where $\mathbf{e}$ is the penultimate column of $\mathbf{E}$. It follows from the Cauchy-Schwartz inequality and Equation (2) that,

$$|\langle \mathbf{t}_x, \mathbf{e} \rangle| \leq \sqrt{\|\mathbf{x}\|^2 + 1} \cdot \|\mathbf{e}\| \leq \sqrt{s^2 \cdot m + 1} \cdot D < q/8 \quad (3)$$

Note that $\mathbf{AS} + \mathbf{E}$ is the concatenation of k vectors $\mathbf{As}_i + \mathbf{e}_i$ for $i \in [k]$. Step 3 is successful if the invert procedure returns $\mathbf{e}_i$ for all i . By Lemma 2.7, this is the case if $\|\mathbf{e}_i\| < q/4\sqrt{s_1(\mathbf{R})^2 + 1}$. From Equation (2), we have

$$\|\mathbf{e}_i\| < D \leq \frac{q}{4B} \leq \frac{q}{4\sqrt{s_1(\mathbf{R})^2 + 1}} \quad (4)$$

It follows from Equations 3 and 4, that for $\mathbf{E}$ such that $\|\mathbf{E}\| < D$, the decryption algorithm successfully returns μ with overwhelming probability over the randomness of the key generation algorithm.

For the second statement, let $\mathbf{C} = \mathbf{AS} + \mathbf{E} + \mu\mathbf{G} \leftarrow \text{GG-GSW.Enc}(\text{pk}, \mu)$. We consider the distribution of a vector $\mathbf{e} = [\mathbf{e}_1; \mathbf{e}_2; e_3]$ where $\mathbf{e}_1 \leftarrow \mathcal{D}_{\mathbb{Z}^m, \alpha}$, $\mathbf{e}_2 \leftarrow \mathcal{D}_{\mathbb{Z}^m, s\hat{\mathbf{E}}\mathbf{T}}$, $e_3 \leftarrow \mathcal{D}_{\mathbb{Z}, \alpha}$, where $\hat{\mathbf{E}} = [\mathbf{E}_1 | \mathbf{I}_m]$. Note that $\mathbf{E}$ is the concatenation of k such vectors. We have that

$$\|(\mathbf{e}_1, e_3)\| \leq \alpha\sqrt{m+1}, \quad \|\mathbf{e}_2\| \leq s \cdot s_1(\hat{\mathbf{E}})\sqrt{m}. \quad (5)$$

except with negligible probability, where the first inequality comes from Lemma 2.3 and the second from Lemma 2.6 with $c = 1$ and Lemma 2.5. We then note that

$$\|e\| = \sqrt{\|\mathbf{e}_1\|e_3\|^2 + \|\mathbf{e}_2\|^2} \quad (6)$$

$$\leq \sqrt{\alpha^2(m+1) + s^2 \cdot s_1(\hat{\mathbf{E}})^2 m} \quad (7)$$

$$\leq \sqrt{\alpha^2(m+1) + s^2 \cdot (C \cdot \alpha(\sqrt{k} + \sqrt{m} + \sqrt{\lambda}) + 1)^2 m} \quad (8)$$

where the last inequality hold with overwhelming probability from Lemma 2.5. From the first statement of the lemma and a union bound, we know that the ciphertext $\mathbf{C}$ correctly decrypts if the term in 8 is less than D . Noticing that $C \cdot \alpha(\sqrt{k} + \sqrt{m} + \sqrt{\lambda}) + 1 = \Theta(\alpha\sqrt{k})$, it is sufficient that

$$\begin{aligned} \frac{q}{8\sqrt{s^2 \cdot m + 1}} &> \sqrt{\alpha^2(m+1) + s^2 \cdot \Theta(\alpha^2 k)m} \\ q &> 8\sqrt{(s^2 m + 1)(\alpha^2(m+1) + s^2 \Theta(\alpha^2 k)m)} \\ q &= \Omega\left(s^2 \alpha m \sqrt{k}\right) \end{aligned}$$

□

Homomorphic Operations The main difference between our scheme and the regular GSW scheme is in the key generation. Hence, once our trapdoored matrix $\mathbf{A}$ is generated, our scheme behaves as the GSW scheme. In particular, it inherits all the homomorphic properties from it. We define the homomorphic operations in exactly the same way as in [6], and show that the noise growth behavior remains unchanged.

- $\text{Add}(\mathbf{C}_1, \mathbf{C}_2)$: $\mathbf{C}_1 \boxplus \mathbf{C}_2 = \mathbf{C}_1 + \mathbf{C}_2$,
- $\text{Mult}(\mathbf{C}_1, \mathbf{C}_2)$: $\mathbf{C}_1 \boxdot \mathbf{C}_2 \leftarrow \mathbf{C}_1 \cdot \mathbf{G}^{-1}(\mathbf{C}_2)$. As in the original GSW scheme, this operation is non-commutative and right associative. A scalar-ciphertext product between a and $\mathbf{C}$ is computed as $\mathbf{C} \boxdot a \leftarrow \mathbf{C} \cdot \mathbf{G}^{-1}(a\mathbf{G})$.

Lemma 4.2. *Let $\mathbf{C}_1, \mathbf{C}_2$ be two ciphertexts encrypting μ_1, μ_2 with noise $\mathbf{E}^{(1)}, \mathbf{E}^{(2)}$ respectively. Then $\mathbf{C}_1 \boxplus \mathbf{C}_2$ is an encryption of $\mu_1 + \mu_2$ and has noise $\mathbf{E}^{(1)} + \mathbf{E}^{(2)}$; $\mathbf{C}_1 \boxdot \mathbf{C}_2$ is an encryption of $\mu_1 \cdot \mu_2$ and has noise $\mathbf{E}^{(1)}\mathbf{X} + \mu_1\mathbf{E}^{(2)}$, where $\mathbf{X} \leftarrow \mathbf{G}^{-1}(\mathbf{C}_2)$.*

The proof is essentially unchanged from [6, Lemma 3.5]. We write it down for completeness.

Proof. We have $\mathbf{C}_i = \mathbf{A}\mathbf{S}_i + \mathbf{E}^{(i)} + \mu_i\mathbf{G}$. The first statement follows from linearity. For the second statement, let $\mathbf{t}_\mathbf{x}$ and $\mathbf{T}_\mathbf{R}$ be defined as in Step 1 from the decryption algorithm. We have,

$$\begin{aligned} \mathbf{t}_\mathbf{x}^\top(\mathbf{C}_1 \boxdot \mathbf{C}_2) &= \mathbf{t}_\mathbf{x}^\top(\mathbf{A}\mathbf{S}_1 + \mathbf{E}^{(1)} + \mu_1\mathbf{G})\mathbf{X} \\ &= \mathbf{t}_\mathbf{x}^\top\mathbf{E}^{(1)}\mathbf{X} + \mu_1\mathbf{t}_\mathbf{x}^\top(\mathbf{A}\mathbf{S}_2 + \mathbf{E}^{(2)} + \mu_2\mathbf{G}) \\ &= \mathbf{t}_\mathbf{x}^\top(\mathbf{E}^{(1)}\mathbf{X} + \mu_1\mathbf{E}^{(2)}) + \mu_1\mu_2\mathbf{t}_\mathbf{x}^\top\mathbf{G} \end{aligned}$$

$$\begin{aligned} \mathbf{T}_\mathbf{R}((\mathbf{C}_1 \boxdot \mathbf{C}_2) - \mu_1\mu_2\mathbf{G}) &= \mathbf{T}_\mathbf{R}(\mathbf{A}\mathbf{S}_1 + \mathbf{E}^{(1)} + \mu_1\mathbf{G})\mathbf{X} - \mu_1\mu_2\mathbf{G} \\ &= \hat{\mathbf{G}}(\mathbf{S}_1\mathbf{X} + \mu_1\mathbf{S}_2) + \mathbf{T}_\mathbf{R}(\mathbf{E}^{(1)}\mathbf{X} + \mu_1\mathbf{E}^{(2)}). \end{aligned}$$

□

4.3 CCA1 Security

Theorem 4.1. *The above scheme is IND-CCA1-secure under the LWE assumption. More precisely, for any PPT adversary $\mathcal{A}$ against the IND-CCA1 security of the scheme, there exist an LWE distinguisher $\mathcal{B}$ such that*

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{IND-CCA1}}(\lambda) \leq \text{Adv}_{\mathcal{B}}^{\text{LWE}}(\lambda) + \text{negl}(\lambda).$$

We first give an intuition on why the scheme is CCA1-secure. As in [34], the condition on the norm of $\mathbf{E}$ in Step 4 is set in such a way to reject all ciphertexts with a noise that would make the `Invert` procedure fail. This allows us

to argue that the output of the decryption procedure is statistically independent from the secret key. We leverage this fact to introduce a second decryption method, based on a statistical search-LWE oracle, which does not need a gadget matrix embedded in the public-key. The crux of the proof is that we can freely switch back and forth between decryption methods, since in the IND-CCA1 game, the adversary does not get access to the decryption oracle anymore once it receives the challenge ciphertext. This allows us to first switch to the oracle-aided decryption method to establish the linear relationship between $\mathbf{C}_1$ and $\mathbf{C}_2$ without using a puncturing argument, since we can still decrypt without the gadget $\hat{\mathbf{G}}$. We then switch back to the regular decryption algorithm to be able to use the LWE assumption.

Important note. The following proof of Theorem 4.1 is incorrect, therefore the claims of IND-CCA1 security are currently unsupported. It is possible for a statistical adversary to distinguish between game G_3 and G_4 : a statistical adversary can compute $\mathbf{S}$ from $(\mathbf{A}_1, \mathbf{C}_1 = \mathbf{A}_1\mathbf{S} + \mathbf{E}_1)$ and with overwhelming probability, $\|\mathbf{C}_2 - \mathbf{A}_2\mathbf{S}\|$ is small in game G_3 but big in game G_4 . We thank an anonymous CRYPTO reviewer for pointing this out.

Proof. We first describe an alternate decryption algorithm that uses a statistical search-LWE oracle instead of the secret-key $\mathbf{sk} = (\mathbf{R}, \mathbf{x})$ to decrypt. The oracle $\mathcal{O}_{\mathbf{A}}(\cdot)$ is parametrized by the public-key matrix $\mathbf{A}$ and the bound D . On input a matrix $\mathbf{B} \in \mathbb{Z}_q^{m \times k}$, $\mathcal{O}_{\mathbf{A}}(\mathbf{B})$ outputs the tuple $(\mathbf{S}, \mathbf{E})$ if there exists $\mathbf{S} \in \mathbb{Z}_q^{n \times k}$, $\mathbf{E} \in \mathbb{Z}_q^{m \times k}$ such that $\mathbf{B} = \mathbf{AS} + \mathbf{E} \bmod q$ and $\|\mathbf{E}\| < D$. Otherwise it outputs $\perp$. Recall that for relevant parameters, the LWE function is injective so the oracle is well-defined. We describe a decryption algorithm $\text{Dec}^{\mathcal{O}_{\mathbf{A}}(\cdot)}$ that has oracle access to $\mathcal{O}_{\mathbf{A}}(\cdot)$ in Figure 2. In Lemma 4.3, we show the following claim.

$\text{Dec}^{\mathcal{O}_{\mathbf{A}}(\cdot)}(\mathbf{C})$:

```

1 :  for  $\mu \in \mathcal{M}$ 
2 :    if  $\mathcal{O}_{\mathbf{A}}(\mathbf{C} - \mu\mathbf{G}) \neq \perp$ 
3 :      then return  $\mu$ 
4 :  return  $\perp$ 

```

Fig. 2. The oracle-aided decryption algorithm $\text{Dec}^{\mathcal{O}_{\mathbf{A}}(\cdot)}(\cdot)$.

Claim. Let $(\mathbf{pk}, \mathbf{sk}) \leftarrow \text{GG-GSW.Gen}(1^\lambda)$. For all $\mathbf{C} \in \mathbb{Z}_q^{m \times k}$, we have that

$$\Pr[\text{GG-GSW.Dec}(\mathbf{sk}, \mathbf{C}) \neq \text{Dec}^{\mathcal{O}_{\mathbf{A}}(\cdot)}(\mathbf{C})] \leq \text{negl}(\lambda)$$

with the probability going over the randomness of the key generation algorithm.

We consider a sequence of hybrid games. For each index i , let W_i be the event that the adversary $\mathcal{A}$ wins in game G_i .

Game G_0 : This is the normal IND-CCA1 game.

Game G_1 : In this game, we change how decryption queries are answered. Upon receiving a decryption query, the challenger uses the oracle-aided decryption algorithm $\text{Dec}^{\mathcal{O}_A(\cdot)}$ instead of GG-GSW.Dec . It follows from Lemma 4.3 that,

$$|\Pr[W_0] - \Pr[W_1]| \leq \text{negl}(\lambda).$$

Game G_2 : We now change how the public key is computed. Sample $\mathbf{A}_1 \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$ and also $\mathbf{a}_3 \xleftarrow{\$} \mathbb{Z}_q^n$ directly from the uniform distribution. Compute $\mathbf{A}_2 = -\mathbf{R}\mathbf{A}_1$ and set $\mathbf{A} = [\mathbf{A}_1; \mathbf{A}_2; \mathbf{a}_3^\top]$ as usual.

Notice that since decryption queries are now answered through the statistical search-LWE oracle, it is not an issue that $\mathbf{A}_2$ does not contain the $\hat{\mathbf{G}}$ term. Hence, it follows from the leftover hash lemma that,

$$|\Pr[W_1] - \Pr[W_2]| \leq \text{negl}(\lambda).$$

Game G_3 : In this game, we change how the challenge ciphertext is computed.

Sample $\mathbf{S} \xleftarrow{\$} \mathbb{Z}_q^{n \times k}$, $\mathbf{E}_1 \leftarrow \mathcal{D}_{\mathbb{Z}^m, \alpha}^k$ and $\mathbf{e}_3 \leftarrow \mathcal{D}_{\mathbb{Z}, \alpha}^k$ as usual then let $\mathbf{C}_1 = \mathbf{A}_1\mathbf{S} + \mathbf{E}_1 \bmod q$ and $\mathbf{c}_3^\top = \mathbf{a}_3^\top\mathbf{S} + \mathbf{e}_3^\top \bmod q$. Let,

$$\mathbf{C}_2 = -\mathbf{R}\mathbf{C}_1 + \bar{\mathbf{E}}_2 \bmod q$$

where $\bar{\mathbf{E}}_2 \leftarrow \mathcal{D}_{\mathbb{Z}^m, s}^k$. Set $\mathbf{C} = [\mathbf{C}_1; \mathbf{C}_2; \mathbf{c}_3^\top]$, the vertical concatenation of $\mathbf{C}_1, \mathbf{C}_2$ and $\mathbf{c}_3^\top$. The challenge ciphertext is then $\mathbf{C}^* = \mathbf{C} + \mu_b\mathbf{G}$. Notice that,

$$\mathbf{C}_2 = -\mathbf{R}\mathbf{A}_1\mathbf{S} - \mathbf{R}\mathbf{E}_1 + \bar{\mathbf{E}}_2 \bmod q.$$

We show that the distribution of $-\mathbf{R}\mathbf{E}_1 + \bar{\mathbf{E}}_2$ conditioned on the view of the adversary is statistically close to $\mathcal{D}_{\mathbb{Z}^m, s\hat{\mathbf{E}}}^k$, the distribution of $\mathbf{E}_2$ in game G_2 . Recall that decryption queries are answered with $\text{Dec}^{\mathcal{O}_A(\cdot)}$ and are independent of $\mathbf{R}$. It is therefore sufficient to consider the distribution of $\mathbf{r}_i^\top\mathbf{E}_1 + \bar{\mathbf{e}}_i^\top$ conditioned on the public-key, where r_i and e_i are the i -th rows of $\mathbf{R}$ and $\bar{\mathbf{e}}_2$. Fixing pk , we have that r_i is sampled from a coset of $\Lambda_q^\perp(\mathbf{A}_1^\top)$. It follows from Lemma C.4 and Lemma C.3 that the distribution of $-\mathbf{r}_i^\top\mathbf{E}_1 + \bar{\mathbf{e}}_i^\top$ is negligibly close to $\mathcal{D}_{\mathbb{Z}^m, s\hat{\mathbf{E}}\tau}$ where $\hat{\mathbf{E}} = [\mathbf{E}_1 | \mathbf{I}_m]$. Finally, from a union bound we have,

$$|\Pr[W_2] - \Pr[W_3]| \leq \text{negl}(\lambda).$$

Game G_4 : In this game, we switch back to the trapdoor-based decryption. We change the public-key again to be $\text{pk} = [\mathbf{A}_1; \hat{\mathbf{G}} - \mathbf{R}\mathbf{A}_1; -\mathbf{x}\mathbf{A}_1]$, and use the original decryption algorithm GG-GSW.Dec to answer decryption queries. It follows from the same arguments as in the hops from G_0 to G_2 that,

$$|\Pr[W_3] - \Pr[W_4]| \leq \text{negl}(\lambda).$$

Game G_5 : In this game, $\mathbf{C}_1 \xleftarrow{\$} \mathbb{Z}_q^{m \times k}$ and $\mathbf{c}_3 \xleftarrow{\$} \mathbb{Z}_q^k$ are now sampled from the uniform distribution. We have that game G_4 and G_5 are computationally indistinguishable under the LWE assumption, from a standard reduction where we use the LWE challenge to simulate $\mathbf{C}_1$ and $\mathbf{c}_3$.

$$|\Pr[W_4] - \Pr[W_5]| \leq \text{Adv}_{\mathcal{B}}^{\text{LWE}}(\lambda).$$

Game G_6 : Finally, in game G_6 we return to using the oracle-aided decryption algorithm $\text{Dec}^{\mathcal{O}_{\mathbf{A}}(\cdot)}$ to answer decryption queries. It follows from Lemma 4.3 that,

$$|\Pr[W_5] - \Pr[W_6]| \leq \text{negl}(\lambda).$$

Let us consider the view of the adversary in game G_6 . The public key and the challenge ciphertext are

$$\text{pk} = \begin{bmatrix} \mathbf{A}_1 \\ \hat{\mathbf{G}} - \mathbf{R}\mathbf{A}_1 \\ -\mathbf{x}\mathbf{A}_1 \end{bmatrix}, \quad \mathbf{C}^* = \begin{bmatrix} \mathbf{C}_1 \\ -\mathbf{R}\mathbf{C}_1 + \bar{\mathbf{E}}_2 \\ \mathbf{c}_3 \end{bmatrix} + \mu_b \mathbf{G}.$$

The challenger uses the oracle-aided $\text{Dec}^{\mathcal{O}_{\mathbf{A}}(\cdot)}$ algorithm to answer decryption queries. This is statistically independent from $\mathbf{R}$ and $\mathbf{x}$. Therefore, it follows from the leftover hash lemma that the distribution of $(\mathbf{A}_1, \mathbf{R}\mathbf{A}_1, \mathbf{x}\mathbf{A}_1, \mathbf{C}_1, \mathbf{R}\mathbf{C}_1, \mathbf{c}_3)$ is statistically close to the uniform distribution. Thus, the advantage of the adversary in Game G_6 is negligible,

$$\Pr[W_6] \leq \frac{1}{2} + \text{negl}(\lambda).$$

□

Lemma 4.3. *Let $(\text{pk} = \mathbf{A}, \text{sk} = (\mathbf{R}, \mathbf{x})) \leftarrow \text{GG-GSW.Gen}(1^\lambda)$. For every $\mathbf{C} \in \mathbb{Z}_q^{m \times k}$,*

$$\Pr[\text{GG-GSW.Dec}(\text{sk}, \mathbf{C}) \neq \text{Dec}^{\mathcal{O}_{\mathbf{A}}(\cdot)}(\mathbf{C})] \leq \text{negl}(\lambda)$$

with the probability going over the randomness of the key generation algorithm.

Proof. For any $\mathbf{C} \in \mathbb{Z}_q^{m \times k}$, it follows from the injectivity of the LWE function that there exists at most one unique tuple $(\mathbf{S}, \mathbf{E}, \mu) \in (\mathbb{Z}_q^{n \times k} \times \mathbb{Z}_q^{m \times k} \times \mathbb{Z}_q)$ such that $\mathbf{C} = \mathbf{A}\mathbf{S} + \mathbf{E} + \mu\mathbf{G}$ with $\|\mathbf{E}\| < D$. If no such tuple exists, both $\text{GG-GSW.Dec}(\text{sk}, \mathbf{C})$ and $\text{Dec}^{\mathcal{O}_{\mathbf{A}}(\cdot)}(\mathbf{C})$ return $\perp$.

If $\mu \leftarrow \text{Dec}^{\mathcal{O}_{\mathbf{A}}(\cdot)}(\mathbf{C})$ with $\mu \neq \perp$, then there exists a tuple $(\mathbf{S}, \mathbf{E}) \in \mathbb{Z}_q^{n \times k} \times \mathbb{Z}_q^{m \times k}$ such that $\mathbf{C} - \mu\mathbf{G} = \mathbf{A}\mathbf{S} + \mathbf{E}$ and $\|\mathbf{E}\| < D$. Let *bad* be the event that Equation (2) does not hold. It follows from Lemma 4.1 that $\text{GG-GSW.Dec}(\text{sk}, \mathbf{C})$ returns μ as well, unless the event *bad* happens.

Conversely, if $\mu \leftarrow \text{GG-GSW.Dec}(\text{sk}, \mathbf{C})$, then it follows from Step 3 and 4 of the decryption algorithm that there exists $(\mathbf{S}, \mathbf{E}) \in \mathbb{Z}_q^{n \times k} \times \mathbb{Z}^{\bar{m} \times k}$ such that $\mathbf{C} = \mathbf{AS} + \mathbf{E} + \mu \mathbf{G}$ and $\|\mathbf{E}\| < D$. Since there exists at most one tuple that fulfills this relationship, $\text{Dec}^{\mathcal{O}_{\mathbf{A}}(\cdot)}(\mathbf{C})$ returns μ too. We have,

$$\Pr[\text{GG-GSW.Dec}(\text{sk}, \mathbf{C}) \neq \text{Dec}^{\mathcal{O}_{\mathbf{A}}(\cdot)}(\mathbf{C})] \leq \text{bad} \leq \text{negl}(\lambda)$$

□

4.4 A Time Space Tradeoff

It is possible to decrypt without $\mathbf{x}$, using only $\mathbf{R}$. The decryption algorithm can run Step 3 and 4 for both $\mathbf{C} - 0\mathbf{G}$ and $\mathbf{C} - 1\mathbf{G}$, and retain the message $\mu \in \{0, 1\}$ for which Step 4 is successful. In fact, this is possible for any polynomially-sized message space $\mathcal{M} \subset \mathbb{Z}_q$, although the time-space tradeoff worsens.

In more detail, the scheme is modified as follows. Set $\bar{m} = 2m$ and $k = \bar{m} \cdot l$. Let $D' < \frac{q}{4B}$ with $B = C \cdot s \cdot (2\sqrt{m} + \sqrt{\lambda}) + 1$ where C is the constant from Lemma 2.5. Let $\mathbf{G} = \mathbf{I}_{2m} \otimes \mathbf{g}^\top \in \mathbb{Z}_q^{\bar{m} \times \bar{m} \cdot l}$.

- $\text{Gen}(1^\lambda)$: Sample $\mathbf{A}_1 \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$ and $\mathbf{R} \leftarrow \mathcal{D}_{\mathbb{Z}, s}^{m \times m}$. Compute $\mathbf{A}_2 = \hat{\mathbf{G}} - \mathbf{RA}_1 \bmod q$. Return the public key/secret key pair,

$$\text{pk} = \mathbf{A} := \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{bmatrix} \in \mathbb{Z}_q^{\bar{m} \times k} \quad \text{sk} := \mathbf{R}.$$

- $\text{Enc}(\text{pk} = \mathbf{A}, \mu \in \{0, 1\})$: Sample $\mathbf{S} \xleftarrow{\$} \mathbb{Z}_q^{n \times k}$, $\mathbf{E}_1 \leftarrow \mathcal{D}_{\mathbb{Z}, \alpha}^{m \times k}$, $\mathbf{E}_2 \leftarrow \mathcal{D}_{\mathbb{Z}, \alpha \hat{\mathbf{E}}^\top}^{m \times k}$, with $\hat{\mathbf{E}} = [\mathbf{E}_1 | \mathbf{I}_m]$ and set $\mathbf{E} = [\mathbf{E}_1; \mathbf{E}_2]$. Return,

$$\mathbf{C} = \mathbf{AS} + \mathbf{E} + \mu \mathbf{G} \bmod q.$$

- $\text{Dec}(\text{sk}, \mathbf{C})$:
 1. Set $\mathbf{T}_{\mathbf{R}} = [\mathbf{R} | \mathbf{I}_m]$. For $b \in \{0, 1\}$, call $\text{Invert}(\mathbf{T}_{\mathbf{R}}, \mathbf{A}, \mathbf{C} - b\mathbf{G})$ to recover values $(\mathbf{S}^{(b)}, \mathbf{E}^{(b)})$. If there is not at least one tuple $(\mathbf{S}^{(b)}, \mathbf{E}^{(b)}) \neq \perp$, return $\perp$.
 2. For $b \in \{0, 1\}$, if $\mathbf{C} = \mathbf{AS}^{(b)} + \mathbf{E}^{(b)} + b\mathbf{G} \bmod q$, set $\mu = b$ and $\mathbf{E} = \mathbf{E}^{(b)}$. If this step fails, return $\perp$.
 3. If $\|\mathbf{E}\| \geq D'$ return $\perp$.
 4. Return μ .

It is clear that both correctness and CCA1 security are preserved. Since we do not need $\mathbf{x}$ to decrypt anymore, this allows us to use $\mathbf{A} = [\mathbf{A}_1; \mathbf{A}_2] \in \mathbb{Z}_q^{2m \times n}$. In turn, this implies that the ciphertext is now a matrix of size $4m^2 \cdot l$ instead of $(2m+1)^2 \cdot l$, and that we tolerate a noise $\mathbf{E}$ with norm bounded by $q/4B$ instead of $q/8s\sqrt{m}$.

5 Achieving IND-CCA1^D Security

We now consider how to achieve IND-CCA1^D security. Since GSW is an exact FHE scheme, we can set the parameters to obtain statistical correctness and obtain IND-CCA1^D security through Lemma 3.3 and Theorem 4.1.

However, this is somewhat unsatisfactory. Most applications do not require such high correctness guarantees in practice, and would benefit more from trading off correctness for better performance. In this section, we show how to obtain IND-CCA1^D secure GSW under parameters that do not provide statistical correctness.

5.1 Rerandomizing Dual-GSW

Our randomized evaluation algorithm requires two components. A first noise-flooding term $\tilde{Y}$ that decouples the rerandomized ciphertexts from the input ciphertexts. This will be chosen according to the HintLWE problem. The second component ensures that the noise remains independent of the message during multiplication. This will use the GSW rerandomization technique introduced by Bourse et al.[8] generalized for dual-GSW.

Note that the results of this section can also be used to obtain a CPA^D-secure scheme from the CPA-secure dual-GSW scheme. Indeed, the results do not depend on the public-key $\mathbf{A}$ or the decryption procedure at all. In the rest of the paper, $\mathbf{G}_{\text{rand}}^{-1}(\cdot)$ is the procedure defined in Definition 2.4.

Lemma 5.1 (adapted from [8], Lemma 3.1). *Let $\varepsilon, \varepsilon' > 0$, and let $r > \eta_\varepsilon(\Lambda_q^\perp(\mathbf{G}))$ be a Gaussian parameter. For any full rank matrix $\mathbf{E} \in \mathbb{Z}_q^{m \times k}$, $\mathbf{v} \in \mathbb{Z}_q^m$, if $r \geq \max\left(8((1-\varepsilon)(2\varepsilon')^2)^{-\frac{1}{m}}, \sqrt{5}(1+s_1(\mathbf{E}))\sqrt{\frac{\ln(2k(1+1/\varepsilon))}{\pi}}\right)$*

$$\Delta((\mathbf{S}, \mathbf{S}\mathbf{x}, \mathbf{E}\mathbf{x} + \mathbf{y}), (\mathbf{S}, \mathbf{u}, \mathbf{e}')) < \varepsilon' + 2\varepsilon$$

where $\mathbf{x} \leftarrow \mathbf{G}_{\text{rand}}^{-1}(v)$ has parameter r , $\mathbf{S} \xleftarrow{\$} \mathbb{Z}_q^{n \times k}$, $\mathbf{u} \xleftarrow{\$} \mathbb{Z}_q^k$, $\mathbf{y} \leftarrow \mathcal{D}_{\mathbb{Z}^m, r}$ and $\mathbf{e}' \leftarrow \mathcal{D}_{\mathbb{Z}^m, r\hat{\mathbf{E}}\tau}$ with $\hat{\mathbf{E}} = [\mathbf{E} | \mathbf{I}_m]$.

Proof. The proof is unchanged from [8, Lemma 3.1]. We simply replace the use of [8, Lemma 3.6] with its matrix generalization Corollary C.1. Using matrix $\mathbf{E}$ instead of vector $\mathbf{e}$ also introduces a factor 2 in the first term of the bound through the generalized leftover hash lemma 2.4.

In the rest of the paper, we overload the notation and write $\text{GG-GSW.Enc}_\Gamma(\text{pk}, x)$ to denote using the encryption algorithm with an ellipsoidal discrete Gaussian noise distribution with parameter Γ .

Lemma 5.2 (adapted from [8], Lemma 4.1). *Let $r = \Omega(\alpha sk \sqrt{\lambda \log k})$ with α and s the Gaussian parameters for the encryption and the key generation respectively. Let $(\text{pk} = \mathbf{A}, \text{sk}) \leftarrow \text{GG-GSW.Gen}(1^\lambda)$. Let $\mathbf{C} = \mathbf{A}\mathbf{S} + \mathbf{E} \leftarrow \text{GG-GSW.Enc}(\text{pk}, 0)$. For any $\mathbf{V} \in \mathbb{Z}_q^{m \times k}$, we have*

$$\left(\mathbf{C} \cdot \mathbf{G}_{\text{rand}}^{-1}(\mathbf{V}) + \mathbf{Y}, \mathbf{C} \right) \stackrel{s}{\approx} (\mathbf{C}', \mathbf{C})$$

where $\mathbf{G}_{\text{rand}}^{-1}(\cdot)$ has parameter r , $\mathbf{Y} \leftarrow \mathcal{D}_{\mathbb{Z}^{\tilde{m} \times k}, r}$, and $\mathbf{C}' \leftarrow \text{GG-GSW.Enc}_\Gamma(\text{pk}, 0)$ with $\Gamma = r[\mathbf{E}|\mathbf{I}_m]$.

Notice that $r = \Omega(s_1(\mathbf{E})\sqrt{\lambda \log k})$ is sufficient for Lemma 5.1. It follows from Lemmas 2.5 and 2.6 that with overwhelming probability ,

$$\begin{aligned} s_1(\mathbf{E}) &\leq s_1(\mathbf{E}_1) + s_1(\mathbf{E}_2) + s_1(\mathbf{e}_3) \\ &\leq \Theta(\alpha \cdot (\sqrt{k} + \sqrt{m} + \sqrt{\lambda})) + \Theta(s \cdot s_1(\mathbf{E}_1)\sqrt{nl}) \\ &= \Theta(\alpha s \sqrt{k \cdot nl}) \end{aligned}$$

Therefore we can choose $r = \Omega(\alpha s k \sqrt{\lambda \log k})$. The proof of Lemma 5.2 then simply follows from replacing with Lemma 5.1 the corresponding lemma in the proof of [8, Lemma 4.1]. We also get the following corollary by noting that $s_1(\mathbf{E} + \tilde{\mathbf{Y}}) \leq s_1(\mathbf{E}) + s_1(\tilde{\mathbf{Y}})$.

Corollary 5.1. *Let $r_1 > 0$ and $r_2 = \Omega((\alpha s k + r_1)\sqrt{\lambda \log(k)})$. Let $\mathbf{A}, \mathbf{C}$ and $\mathbf{V}$ be defined as above. We have,*

$$\left((\mathbf{C} + \tilde{\mathbf{Y}}) \cdot \mathbf{G}_{\text{rand}}^{-1}(\mathbf{V}) + \mathbf{Y}, \mathbf{C} \right) \stackrel{s}{\approx} (\mathbf{C}', \mathbf{C})$$

where $\tilde{\mathbf{Y}} \leftarrow \mathcal{D}_{\mathbb{Z}^{\tilde{m} \times k}, r_1}$, $\mathbf{G}_{\text{rand}}^{-1}(\cdot)$ is parametrized with r_2 , $\mathbf{Y} \leftarrow \mathcal{D}_{\mathbb{Z}^{\tilde{m} \times k}, r_2}$ and $\mathbf{C}' \leftarrow \text{GG-GSW.Enc}_\Gamma(\text{pk}, 0)$ with $\Gamma = r_2[\mathbf{E} + \tilde{\mathbf{Y}}|\mathbf{I}_m]$.

5.2 GG-GSW For Branching Programs

Let us recall the technique introduced by Brakerski and Vaikuntanathan [12] to homomorphically evaluate branching programs. Then, we will introduce our randomized evaluation algorithm.

Permutation branching programs. A permutation branching program Π of length L and width W with input space $\{0, 1\}^l$ is a sequence of L tuples of the form $(\text{var}(t), \pi_{t,0}, \pi_{t,1})$, where

- $\text{var}(t) : [L] \rightarrow [l]$ is a function that associates the tuple for the t -th step to an input bit $x_{\text{var}(t)}$.
- $\pi_{t,0}, \pi_{t,1} : [W] \rightarrow [W]$ are permutations associated to the t -th step of the program that dictate the change of the state as a function of an input bit $x_{\text{var}(t)}$.

On input a tuple $(x_1, \dots, x_l) \in \{0, 1\}^l$, the program Π proceeds as follows. At every period $t \in [L]$, the program has a state $\mathbf{v}_t \in \{0, 1\}^W$, with the initial state $\mathbf{v}_0 = (1, 0, \dots, 0)$. Then, for every $t \in L$ and $w \in [W]$, the state is computed recursively as

$$\mathbf{v}_t[w] = x_{\text{var}(t)} \cdot v_{t-1}[\pi_{t,1}^{-1}(w)] + (1 - x_{\text{var}(t)}) \cdot v_{t-1}[\pi_{t,0}^{-1}(w)]$$

where $\mathbf{v}_t[w]$ is the w -th coefficient of $\mathbf{v}_t$. The output of the program is 1 if $\mathbf{v}_L[0] = 1$ and 0 otherwise.

Randomized homomorphic evaluation of branching programs. Let $(\text{pk} = \mathbf{A}, \text{sk}) \leftarrow \text{GG-GSW.Gen}(1^\lambda)$. The homomorphic evaluation $\text{Eval}(\Pi, (\mathbf{C}_1, \dots, \mathbf{C}_l))$ of a branching program Π on input $(\mathbf{C}_1, \dots, \mathbf{C}_l)$ proceeds as follows. The initial state $\mathbf{V}_0 = (\mathbf{G}, \mathbf{0}, \dots, \mathbf{0})$ is a tuple of noiseless encryptions of $\mathbf{v}_0 = (1, 0, \dots, 0)$. For $i \in [l]$, the evaluation algorithm rerandomizes $\mathbf{C}_i$ and sets,

$$\mathbf{C}'_i = \mathbf{C}_i + \tilde{\mathbf{Y}}_i \quad (9)$$

where $\tilde{\mathbf{Y}}_i \leftarrow \mathcal{D}_{\mathbb{Z}^{\bar{m}} \times k, r_1}$. Then the program is iteratively evaluated as,

$$\mathbf{V}_t[w] = \mathbf{C}'_{\text{var}(t)} \cdot \mathbf{G}_{\text{rand}}^{-1}(\mathbf{V}_{t-1}[\pi_{t,1}^{-1}(w)]) + (\mathbf{G} - \mathbf{C}'_{\text{var}(t)}) \cdot \mathbf{G}_{\text{rand}}^{-1}(\mathbf{V}_{t-1}[\pi_{t,0}^{-1}(w)]) + \mathbf{Y}_{t,w} \quad (10)$$

where $\mathbf{Y}_{t,w} \leftarrow \mathcal{D}_{\mathbb{Z}^{\bar{m}} \times k, r_2 \sqrt{2}}$. The evaluation algorithm outputs the ciphertext $\mathbf{V}_L[0]$.

We now analyse the behavior of our randomized homomorphic evaluation algorithm over branching programs. The following results follow the methodology from Bourse et al. [8] exactly, simply replacing the relevant parts with their matrix equivalent, so we omit the proof.

Lemma 5.3 (adapted from [8], Lemma 5.3). *Let $r_1 > 0$ be a Gaussian parameter and let $r_2 = \Omega((\alpha sk + r_1) \sqrt{\lambda \log(k)})$. Let $\mu, v_0, v_1 \in \{0, 1\}$ and $(\text{pk} = \mathbf{A}, \text{sk}) \leftarrow \text{GG-GSW.Gen}(1^\lambda)$. Let $\mathbf{C} = (\mathbf{A}\mathbf{S} + \mathbf{E} + \mu\mathbf{G}) \leftarrow \text{GG-GSW.Enc}(\text{pk}, x)$ and $\mathbf{C}' = \mathbf{C} + \tilde{\mathbf{Y}}$ with $\tilde{\mathbf{Y}} \leftarrow \mathcal{D}_{\mathbb{Z}^{\bar{m}}, r_1}^k$. We have,*

$$\left(\mathbf{C}' \cdot \mathbf{G}_{\text{rand}}^{-1}(\mathbf{V}_1) + (\mathbf{G} - \mathbf{C}') \cdot \mathbf{G}_{\text{rand}}^{-1}(\mathbf{V}_0) + \mathbf{Y}, \mathbf{C} \right) \stackrel{s}{\approx} (\mathbf{V}'_x, \mathbf{C})$$

where $\mathbf{V}_b \leftarrow \text{GG-GSW.Enc}_\Gamma(\text{pk}, v_b)$ for $b \in \{0, 1\}$, $\mathbf{Y} \leftarrow \mathcal{D}_{\mathbb{Z}^{\bar{m}}, r_2 \sqrt{2}}^k$, and $\mathbf{V}'_x \leftarrow \text{GG-GSW.Enc}_\Xi(\text{pk}, v_x)$ with $\Xi = (\Gamma^\top \Gamma + 2r_2^2 \hat{\mathbf{E}}^\top \hat{\mathbf{E}})^{\frac{1}{2}}$ and $\hat{\mathbf{E}} = [\mathbf{E} + \tilde{\mathbf{Y}} | \mathbf{I}_{\bar{m}}]$.

Informally, this lemma shows that after one step of the evaluation of a branching program, the noise of the output is independent of the message bits, but depends on the noise $\mathbf{E} + \tilde{\mathbf{Y}}$ of the (rerandomized) input ciphertext. Finally, the following lemma covers the induction over the evaluation of the whole branching program.

Lemma 5.4 (adapted from [8], Lemma 5.4). *For a branching program $\Pi : \{0, 1\}^l \rightarrow \{0, 1\}$ of length L , let $\tau_{t,i}$ be the number of times the i -th variable x_i has been called after t steps of the evaluation. Let $\mathbf{pk}, r_1, r_2$ be as in Lemma 5.3. For any $(x_1, \dots, x_l) \in \{0, 1\}^l$, let $\mathbf{C}_i = \mathbf{A}\mathbf{S}_i + \mathbf{E}_i + x_i \mathbf{G} \leftarrow \text{GG-GSW.Enc}(\mathbf{pk}, x_i)$. For all $t \in [L]$ and $w \in [W]$, let $\mathbf{V}_t[w]$ be as defined in Equation (10). We have,*

$$\left(\mathbf{V}_t[w], (\mathbf{C}_i)_{i=1}^l \right) \stackrel{s}{\approx} (\mathbf{V}'_{t,w}, (\mathbf{C}_i)_{i=1}^l)$$

where $\mathbf{V}'_{t,w} \leftarrow \text{GG-GSW.Enc}_{\Gamma_t}(\mathbf{pk}, v_t[w])$ with $\Gamma_t = r_2 \left(2 \sum_{i=1}^l \tau_{t,i} \hat{\mathbf{E}}_i \hat{\mathbf{E}}_i^\top \right)^{1/2}$ and $\hat{\mathbf{E}}_i = [\mathbf{E}_i + \tilde{\mathbf{Y}}_i | \mathbf{I}_m]$.

5.3 IND-CCA1^D Secure GSW Without Correctness Guarantees

We are now ready to show that our randomized homomorphic evaluation algorithm makes GG-GSW an IND-CCA1^D secure leveled FHE scheme for branching programs.

Theorem 5.1. *GG-GSW with the randomized evaluation algorithm for branching programs from Lemma 5.3 achieves IND-CCA1^D security under the HintLWE assumption. More precisely, for any PPT adversary $\mathcal{A}$ that makes $l = l(\lambda)$ encryption queries and $h = h(\lambda)$ homomorphic evaluation queries, there exists a PPT HintLWE distinguisher $\mathcal{B}$ such that,*

$$\text{Adv}_{\text{GG-GSW}, \mathcal{A}}^{\text{IND-CCA1}^D}(\lambda) \leq l \cdot \text{Adv}_{\mathcal{B}}^{\text{HintLWE}_{n,m,q,\alpha}^{k,h,r_1,c}}(\lambda) + \text{negl}(\lambda)$$

Proof. We consider a sequence of games G_i . For every index i , let W_i be the event that the adversary wins the game G_i .

Game G_0 : This is the normal IND-CCA1^D game.

Game G_1 : Similarly to the changes we made in the proof of CCA1 security from Theorem 4.1, the queries to the unrestricted decryption oracle are answered through a statistical search-LWE oracle. Additionally, we change how the public-key is computed. Let $\mathbf{A}_1 \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$ and $\mathbf{a}_3 \xleftarrow{\$} \mathbb{Z}_q^n$. Set $\mathbf{A} = [\mathbf{A}_1; -\mathbf{R}\mathbf{A}_1; \mathbf{a}_3^\top]$ the vertical concatenation of $\mathbf{A}_1$, $-\mathbf{R}\mathbf{A}_1$ and $\mathbf{a}_3^\top$. As in the proof of Theorem 4.1, we have

$$|\Pr[W_0] - \Pr[W_1]| \leq \text{negl}(\lambda).$$

Game G_2 : We now change the CPA^D oracles. Consider the following modifications.

- $\text{E}(m_0, m_1)$: For every encryption query, the oracle returns $\mathbf{C} = \mathbf{A}\mathbf{S} + \mathbf{E} + m_b \mathbf{G} \bmod q$ as usual. When updating the state, it additionally keeps track of the encryption noise $\mathbf{E}$, i.e. $S \leftarrow S \cup (m_0, m_1, \mathbf{C}, \mathbf{E})$.

- $H(\Pi, J)$: Similarly, for every homomorphic evaluation query, the oracle keeps track of the noise in the ciphertext, i.e.

$$S \cup (\Pi(S[j_1].m_0, \dots, S[j_i].m_0), \Pi(S[j_1].m_1, \dots, S[j_i].m_1), \hat{\mathbf{c}}, \hat{\mathbf{E}}),$$

where $\hat{\mathbf{E}}$ is computed from the corresponding $S[j_i].\mathbf{E}$ and the evaluation randomness.

- $D(j)$: If $S[j].m_0 \neq S[j].m_1$, return $\perp$. Else the restricted decryption oracle sets $\mu = S[j].m_0$ and $\mathbf{E} = S[j].\mathbf{E}$ and runs Step 4 and 5 of the decryption algorithm.

In game G_1 , if $S[j].m_0 = S[j].m_1$ the restricted decryption algorithm returns $\text{Dec}_{sk}(S[j].C)$. Remember that if there exists any $\mathbf{S} \in \mathbb{Z}_q^{n \times k}$ and $\mathbf{E}$ with $\|\mathbf{E}\| < D$ such that $S[j].\mathbf{C} = \mathbf{A}\mathbf{S} + \mathbf{E} + S[j].m_0 \bmod q$, Step 3 of the decryption algorithm will return $\mathbf{E}$. Notice that in game G_2 we have $S[j].\mathbf{C} = \mathbf{A}\mathbf{S} + S[j].\mathbf{E} + S[j].m_0 \bmod q$. It follows from the injectivity of the LWE function that $\mathbf{E} = S[j].\mathbf{E}$. Hence the output of the oracles in game G_1 and G_2 are identical, and we have $\Pr[W_1] = \Pr[W_2]$.

Game G_3 : In this game, we change how the encryption oracle answers queries. On input a query (m_0, m_1) , sample $\mathbf{S} \xleftarrow{\$} \mathbb{Z}_q^{n \times k}$, $\mathbf{E}_1 \leftarrow \mathcal{D}_{\mathbb{Z}, \alpha}^{m \times k}$ and $\mathbf{e}_3 \leftarrow \mathcal{D}_{\mathbb{Z}, \alpha}^k$ as usual then let $\mathbf{C}_1 = \mathbf{A}_1\mathbf{S} + \mathbf{E}_1 \bmod q$ and $\mathbf{c}_3^\top = \mathbf{a}_3^\top\mathbf{S} + \mathbf{e}_3^\top \bmod q$. Let,

$$\mathbf{C}_2 = -\mathbf{R}\mathbf{C}_1 + \bar{\mathbf{E}}_2 \bmod q$$

where $\bar{\mathbf{E}}_2 \leftarrow \mathcal{D}_{\mathbb{Z}^{n \cdot l \times k}, s}$. Set $\mathbf{C} = [\mathbf{C}_1; \mathbf{C}_2; \mathbf{c}_3^\top]$, the vertical concatenation of $\mathbf{C}_1, \mathbf{C}_2$ and $\mathbf{c}_3^\top$. The encryption oracle returns $\mathbf{C} = \mathbf{C} + m_b\mathbf{G}$. Let $\mathbf{E}_2 = -\mathbf{R}\mathbf{E}_1 + \bar{\mathbf{E}}_2$ and set $\mathbf{E} = [\mathbf{E}_1; \mathbf{E}_2; \mathbf{e}_3^\top]$. The oracle updates the state with $\mathbf{E}$. As in the proof of Theorem 4.1, we have

$$|\Pr[W_2] - \Pr[W_3]| \leq \text{negl}(\lambda).$$

Game G_4 : In this game, we modify the encryption algorithm as follows. The oracle simply outputs

$$\mathbf{C} = \mathbf{C}' + m_b\mathbf{G}$$

where $\mathbf{C}' \xleftarrow{\$} \mathbb{Z}_q^{\bar{m} \times k}$ is directly sampled from the uniform distribution. The oracle still samples $\mathbf{E} \in \mathbb{Z}^{\bar{m} \times k}$ and updates the state with $\mathbf{C}$ and $\mathbf{E}$ as in game G_3 . We show that game G_3 and G_4 are indistinguishable under the HintLWE assumption.

To this end, we introduce a series of intermediary hybrid games $G_{3,j}$. Without loss of generality, we can assume that an adversary $\mathcal{A}$ makes $l = l(\lambda)$ queries to the encryption oracle and $h = h(\lambda)$ queries to the homomorphic evaluation oracle. Then, for $j \in [l]$ we define:

Game $G_{3,j}$: In this game, the first j encryption queries are answered as in game G_3 . The last $l - j$ encryption queries are answered as in game G_4 . For convenience we define game $G_{3,0}$ to be game G_3 . We show in Lemma 5.5 that

$|\Pr[W_{3,j-1}] - \Pr[W_{3,j}]| \leq \text{Adv}_{\mathcal{B}}^{\text{HintLWE}_{n,m,q,\alpha}^{k,h,r_1,c}}(\lambda)$. It follows from a standard hybrid argument that,

$$|\Pr[W_3] - \Pr[W_4]| \leq l \cdot \text{Adv}_{\mathcal{B}}^{\text{HintLWE}_{n,m,q,\alpha}^{k,h,r_1,c}}(\lambda).$$

Finally, notice that in game G_4 , the key and the ciphertexts are negligibly close to uniformly random and we have,

$$\Pr[W_4] = \frac{1}{2} + \text{negl}(\lambda).$$

□

Lemma 5.5. *For $j \in [l]$, we have that game $G_{3,j-1}$ is computationally indistinguishable from $G_{3,j}$ under the HintLWE assumption. More precisely, for any PPT distinguisher $\mathcal{A}$ that distinguishes between game $G_{3,j-1}$ and game $G_{3,j}$, there exists a PPT distinguisher $\mathcal{B}$ such that $|\Pr[W_{3,j-1}] - \Pr[W_{3,j}]| \leq \text{Adv}_{\mathcal{B}}^{\text{HintLWE}_{n,m,q,\alpha}^{k,h,r_1,c}}(\lambda)$.*

Proof. The distinguisher $\mathcal{B}$ receives a HintLWE challenge tuple

$$((\mathbf{A}', \mathbf{B}'), (1, \mathbf{H}^{(\psi)} = \mathbf{E}' + \tilde{\mathbf{Y}}'^{(\psi)})_{\psi=1}^h),$$

where $\mathbf{A}' = [\mathbf{A}_1; \mathbf{a}_3^T], \mathbf{B} = [\mathbf{B}_1; \mathbf{b}_3^T] \in \mathbb{Z}_q^{m \times k} \times \mathbb{Z}_q^k$, $\mathbf{E}' = [\mathbf{E}_1; \mathbf{e}_3^T] \in \mathbb{Z}_q^{m \times k} \times \mathbb{Z}_q^k$ and $\tilde{\mathbf{Y}}'^{(\psi)} = [\tilde{\mathbf{Y}}_1^{(\psi)}; \tilde{\mathbf{y}}_3^{(\psi)T}] \in \mathbb{Z}^{m \times k} \times \mathbb{Z}^k \leftarrow \mathcal{D}_{\mathbb{Z}^{m+1 \times k}, r_1}$ for $\psi \in [h]$. The task is to distinguish whether $\mathbf{B} = \mathbf{A}'\mathbf{S}' + \mathbf{E}' \bmod q$ for some $\mathbf{S}' \in \mathbb{Z}_q^{n \times k}$ or is uniformly random.

The distinguisher $\mathcal{B}$ samples $\mathbf{R}$ and sets $\mathbf{A} = [\mathbf{A}_1, -\mathbf{R}\mathbf{A}_1, \mathbf{a}_3^T]$. Then it runs $\mathcal{A}(1^\lambda, \mathbf{A})$, initializes the state S and answers queries as follows:

- $\text{E}(m_0, m_1)$: The i -th encryption query is answered as follows.
 - If $i < j$: Sample $\mathbf{S}, \mathbf{E}_1, \mathbf{e}_3$ and compute $\mathbf{C}$ and $\mathbf{E}$ as in game G_3 . Output $\mathbf{C}$ and update the state with $S \cup (m_0, m_1, \mathbf{C}, \mathbf{E})$.
 - If $i > j$: Return $\mathbf{C} = \mathbf{C}' + m_b \mathbf{G}$ with $\mathbf{C}' \xleftarrow{\$} \mathbb{Z}_q^{m \times k}$ and update the state as in game G_4 .
 - If $i = j$: Set $\mathbf{C}_1 = \mathbf{B}_1$ and $\mathbf{c}_3 = \mathbf{b}_3$. Compute $\mathbf{C}_2 = -\mathbf{R}\mathbf{C}_1 + \bar{\mathbf{E}}_2 \bmod q$ with $\bar{\mathbf{E}}_2 \leftarrow \mathcal{D}_{\mathbb{Z}^{n \times k}, r}$. Output $\mathbf{C} = [\mathbf{C}_1; \mathbf{C}_2; \mathbf{c}_3^T] + m_b \mathbf{G}$ and update the state with $S \leftarrow S \cup (m_0, m_1, \mathbf{C}, \perp)$.
- $\text{H}(\Pi, J)$: Without loss of generality, we consider the single-hop case. For the multi-hop case, $\mathcal{B}$ keeps track of the “chain” of homomorphic operations to be able to sample from the right distributions. On input (Π, J) the ψ -th homomorphic evaluation query for $\psi \in [h]$, let $\mathbf{C}_1, \dots, \mathbf{C}_{l'}$ be the ciphertexts corresponding to the tuple of indices J , let $x_1, \dots, x_{l'} \in \{0, 1\}$ be the corresponding messages they encrypt and let $\mathbf{E}_i$ be the corresponding ciphertext noise i.e. $x_i = S[i].m_b$ and $\mathbf{E}_i = S[i].\mathbf{E}$ for $i \in J$. Let $\hat{x} = \Pi(x_1, \dots, x_{l'})$. Set,

$$\hat{\mathbf{E}}_i = \begin{cases} [\mathbf{E}_i + \tilde{\mathbf{Y}}_i | \mathbf{I}_{\bar{m}}] & \text{with } \tilde{Y}_i \leftarrow \mathcal{D}_{\mathbb{Z}^{\bar{m} \times k}, r_1}, \quad \text{if } i \neq j \\ [\mathbf{H}^{(\psi)} | \mathbf{I}_{\bar{m}}] & \text{if } i = j. \end{cases}$$

The evaluated ciphertext is computed as $\hat{\mathbf{C}} \leftarrow \text{GG-GSW.Enc}_{\Gamma_L}(\text{pk}, \hat{x})$ where $\Gamma_L = r_2 \left(2 \sum_{i=1}^l \tau_{L,i} \hat{\mathbf{E}}_i \hat{\mathbf{E}}_i^\top \right)$.

- $\mathbf{D}(j)$: Queries to the restricted decryption oracle are answered as in game G_3 .

Recall that the restricted decryption oracle is unchanged between game G_3 and G_4 . It follows from Lemma 5.4 that the output of $\mathbf{H}$ is correctly distributed. Finally, notice that if $\mathbf{B} = \mathbf{A}'\mathbf{S}' + \mathbf{E}' \bmod q$, then $\mathcal{B}$ perfectly simulates hybrid $G_{3,j-1}$ whereas if $\mathbf{B}$ is uniformly distributed, $\mathcal{B}$ simulates $G_{3,j}$. Hence, $\mathcal{B}$ outputs b' the output of $\mathcal{A}$ and has the same advantage in the HintLWE problem as $\mathcal{A}$. $\square$

Corollary 5.2. *Let $\sigma = \Omega(\sqrt{\lambda})$. Let $\alpha = 2\sigma$ and $r_1 = 2\sigma\sqrt{h}$. Then, GG-GSW with the randomized evaluation algorithm for branching programs from Lemma 5.3 achieves IND-CCA1^D security under the $\text{LWE}_{n,m,q,\sigma}^k$ assumption. More precisely, for any PPT adversary $\mathcal{A}$ that makes $l = l(\lambda)$ encryption queries and $h = h(\lambda)$ homomorphic evaluation queries, there exists a PPT LWE distinguisher $\mathcal{B}$ such that,*

$$\text{Adv}_{\text{GG-GSW}, \mathcal{A}}^{\text{IND-CCA1}^D}(\lambda) \leq l \cdot \text{Adv}_{\mathcal{B}}^{\text{LWE}_{n,m,q,\sigma}^k}(\lambda) + \text{negl}(\lambda)$$

Proof. We have that $2(1/\alpha^2 + h/r_1^2) = 2(1/4\sigma^2 + 1/4\sigma^2) = 1/\sigma^2$. From Theorem 2.1, it follows that if $\sigma \geq \sqrt{2}\eta_\varepsilon(\mathbb{Z})$ for some $0 < \varepsilon \leq 1/2$, we have $\text{Adv}_{\mathcal{B}}^{\text{HintLWE}_{n,m,q,\alpha}^{k,h,r_1,C}}(\lambda) = \text{Adv}_{\mathcal{B}}^{\text{LWE}_{n,m,q,\sigma}^k}(\lambda) + 2mk\varepsilon$. We conclude the proof by setting $\varepsilon \leq 2^{-\lambda}$ and using Lemma 2.1. $\square$

5.4 Parameters

Finally, we highlight a key advantage for parameter selection. Because our proof of IND-CCA1^D security does not rely on correctness, we can fix parameters that ensure LWE hardness while, independently, pushing correctness margins extremely aggressively.

Let us first consider the security requirements. Choose a target LWE problem with parameters n, m, k, q, σ with $n = \text{poly}(\lambda)$, $q = \text{poly}(n)$ some power of two, $m = \Omega(n \log q)$ for the leftover hash lemma, $k = m \log q$ to fit the gadget matrix and $s = \Omega(m\alpha\lambda\sqrt{k})$. We also need $\sigma = \Omega(\sqrt{n})$ and following Corollary 5.2, we set $\alpha = 2\sigma$ and $r_1 = 2\sigma\sqrt{h}$, for some polynomial $h = \text{poly}(n)$ (note that in practice it is possible to choose t the hint budget as $\min(h, d)$, where h is an upper bound on the expected number of homomorphic evaluations and d an upper bound on the number of decryptions). From Lemma 5.4, we also require $r_2 = \Omega((\alpha sk + r_1)\sqrt{\lambda \log(k)})$. Let $L_{\max} = \text{poly}(n)$ be the maximum size of the branching programs we accept.

From the constraints, let us set $n = \Theta(\lambda)$, $s = \tilde{\Theta}(n^3)$, $\sigma = \Theta(\sqrt{n})$, $r_1 = \Theta(\sqrt{hn})$, $r_2 = \tilde{\Theta}(n^5 + n\sqrt{h})$ and $q = \tilde{\Theta}((n^{18} + n^{10}h)L_{\max})$ a power of 2. Note that assuming $n \gg h$, which would be the case for a lot of applications, we have $r_2 = \tilde{\Theta}(n^5)$ and $q = \tilde{\Theta}(n^{18}L_{\max})$.

Lemma 5.6. *Let Π be a branching program of length L . Let $(x_1, \dots, x_l) \in \{0, 1\}^l$ and $\mathbf{C}_i = \text{GG-GSW.Enc}(\text{pk}, x_i)$ for $i \in [l]$. Let $\hat{\mathbf{C}} \leftarrow \text{GG-GSW.Eval}(\Pi, (\mathbf{C}_1, \dots, \mathbf{C}_l))$. With the above parameters, we have $\text{Dec}_{\text{sk}}(\hat{\mathbf{C}}) = \Pi(x_1, \dots, x_l)$ with overwhelming probability.*

Proof. Let $\mathbf{E}_i$ be the noise matrices of $\mathbf{C}_i$. From Lemma 5.4 we have that the noise $\hat{\mathbf{E}}$ of $\hat{\mathbf{C}}$ follows a discrete Gaussian distribution with parameter matrix $\Gamma = r_2 \left(2 \sum_{i=1}^l \tau_{L,i} \hat{\mathbf{E}}_i \hat{\mathbf{E}}_i^\top \right)^{1/2}$ where $\hat{\mathbf{E}}_i = [\mathbf{E}_i + \mathbf{Y}_i | \mathbf{I}_{\bar{m}}]$. We have $s_1(\Gamma) \leq r_2 \sqrt{2 \sum_{i=1}^l \tau_{L,i} (s_1(\mathbf{E}'_i \mathbf{E}'_i^\top) + 1)}$ with $\mathbf{E}'_i = \mathbf{E}_i + \mathbf{Y}_i$.

As discussed in Section 5.1, we have $s_1(\mathbf{E}'_i) \leq \delta$ with $\delta = \Theta(\alpha sk \sqrt{\log n} + r_1)$ for all i , except with negligible probability. Since $\sum_i \tau_{L,i} = L$, we have, $s_1(\Gamma) \leq \sqrt{2} r_2 \cdot L \cdot (\delta^2 + 1)$. It follows from Lemma 2.6 and a union bound that with overwhelming probability, $\|\mathbf{E}\| < s_1(\Gamma) \sqrt{m} \log n < \sqrt{2} r_2 \cdot L \cdot (\delta^2 + 1) \sqrt{m} \log n$.

Finally, Lemma 4.1 gives that $\text{GG-GSW.Dec}(\hat{\mathbf{C}})$ returns $\Pi(x_1, \dots, x_l)$ with overwhelming probability for,

$$\begin{aligned} q &= \tilde{\Theta}(s \sqrt{m} r_2 \cdot L_{\max} \delta^2 \sqrt{m}) \\ &= \tilde{\Theta}((n^{18} + n^{10}h + n^9 \sqrt{h} + n^6 h) L_{\max}) \\ &= \tilde{\Theta}((n^{18} + n^{10}h) L_{\max}) \end{aligned}$$

□

Trading correctness for better performance. Notice that Theorem 5.1 relies on the rerandomization from Corollary 5.1, which itself only needs an upper bound on $\mathbf{E} + \tilde{\mathbf{Y}}$. In particular, IND-CCA1^D security holds for any modulus q that yields a secure LWE instance. Say we decide to trade off evaluation correctness for better performance. In that case, we may simply ignore the above lemma and choose q through a tool like the lattice estimator [5], while using our favorite heuristic to estimate evaluation correctness. Importantly, none of the security guarantees rely on having “good” correctness estimations.

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A Standard Security Notions

We recall the definitions of IND-CPA^D and IND-CCA1 security.

Definition A.1 (IND-CPA^D). *Let $\Pi = (\text{Gen}, \text{Enc}, \text{Dec}, \text{Eval})$ be a public-key HE scheme. Consider the IND-CPA^D experiment given in Figure 3. The advantage of an adversary $\mathcal{A}$ is given by:*

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{IND-CPA}^D}(\lambda) = |\Pr[\text{Exp}_{\Pi, \mathcal{A}}^{\text{IND-CPA}^D}(\lambda) = 1] - \frac{1}{2}|.$$

We say that Π is indistinguishable against chosen-plaintext attacks with passive decryption (IND-CPA^D) if for all PPT adversaries $\mathcal{A}$, $\text{Adv}_{\Pi, \mathcal{A}}^{\text{IND-CPA}^D}(\lambda)$ is negligible in λ .

$\text{Exp}_{H,A}^{\text{IND-CPA}^D}(\lambda)$	$E_b(m_0, m_1)$ with $m_0, m_1 \in \mathcal{M}$
1 : $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$	1 : $c \leftarrow \text{Enc}_{\text{pk}}(m_b)$
2 : $b \xleftarrow{\$} \{0, 1\}, S := \emptyset$	2 : $S \leftarrow S \cup (m_0, m_1, c)$
3 : $b' \leftarrow \mathcal{A}^{E_b(\cdot), H(\cdot), D(\cdot)}(s)$	3 : return c
4 : return $(b \stackrel{?}{=} b')$	
	$D(j)$ with $j \in \{1, \dots, S \}$
	1 : if $S[j].m_0 \neq S[j].m_1$
	2 : then return $\perp$
	3 : else
	4 : return $\text{Dec}_{\text{sk}}(S[j].c)$
$H(f, J)$ with $f \in \mathcal{F}$ and $J = (j_1, \dots, j_l)$	
1 : $\hat{c} \leftarrow \text{Eval}(f, S[j_1].c, \dots, S[j_l].c)$	
2 : $S \leftarrow S \cup (f(S[j_1].m_0, \dots, S[j_l].m_0), f(S[j_1].m_1, \dots, S[j_l].m_1), \hat{c})$	
3 : return $\hat{c}$	

Fig. 3. The IND-CPA^D experiment $\text{Exp}_{H,A}^{\text{IND-CPA}^D}(\lambda)$.

Definition A.2 (IND-CCA1). *Let $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ be a public-key encryption scheme. Consider the IND-CCA1 experiment given in Figure 4. The advantage of an adversary $\mathcal{A}$ is given by:*

$$\text{Adv}_{H,A}^{\text{IND-CCA1}}(\lambda) = |\Pr[\text{Exp}_{H,A}^{\text{IND-CCA1}}(\lambda) = 1] - \frac{1}{2}|.$$

We say that Π is indistinguishable against non-adaptive chosen-ciphertext attacks (IND-CCA1) if for all PPT adversaries $\mathcal{A}$, $\text{Adv}_{H,A}^{\text{IND-CCA1}}(\lambda)$ is negligible in λ .

B Proof of Lemma 3.1

Proof. Let $\Pi' = (\text{Gen}', \text{Enc}', \text{Dec}', \text{Eval}')$ be an exact IND-CCA1 secure FHE scheme. We build an *approximate* FHE scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec}, \text{Eval})$ as follows:

- $\text{Gen}(1^\lambda)$: Return $(\text{pk}, \text{sk}) \leftarrow \text{Gen}'(1^\lambda)$.
- $\text{Enc}_{\text{pk}}(m)$: Return $c = (\text{Enc}'_{\text{pk}}(m), \text{Enc}'_{\text{pk}}(m))$.
- $\text{Eval}(f, c)$: Parse c as (c_0, c_1) . Return $\hat{c} = (\text{Eval}'(f, c_0), c_1)$.

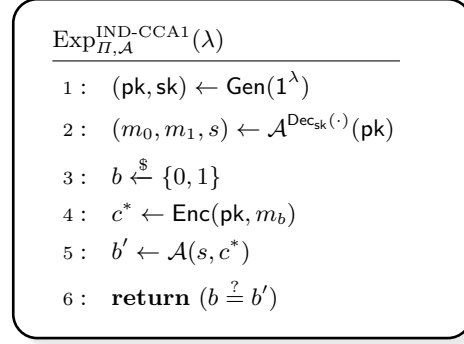


Fig. 4. The IND-CCA1 experiment $\text{Exp}_{\Pi, \mathcal{A}}^{\text{IND-CCA1}}(\lambda)$.

- $\text{Dec}_{\text{sk}}(c)$: Parse c as (c_0, c_1) . Compute $m = \text{Dec}'_{\text{sk}}(c_0)$ and $d = \text{Dec}'_{\text{sk}}(c_1)$. If either m or d is the $\perp$ symbol, return $\perp$. If $d = 0$ return m . Else return $m + \alpha$, where $\alpha \neq 0$ is a fixed public element.

We only give the case of univariate functions f , but the general case follows naturally. Notice that Π is an IND-CCA1 secure approximate FHE scheme. Indeed, since α is public, the reduction can handle the modification of the decryption algorithm by using its IND-CCA1 decryption oracle and adding α when appropriate.

However, there exists an efficient IND-CPA^D attack on Π . Query $(0, B)$ to the encryption oracle for any non-zero B . This adds $(0, B, c)$ to the state. Then query the subtraction of c with itself. This adds $(0, 0, \hat{c})$ to the state of the oracles, where $\hat{c} = (c_0, c_1)$. We can now query $\hat{c}$ (more precisely its index) to the decryption oracle. If $b = 0$, $\text{Dec}_{\text{sk}}(c_1) = 0$, and the oracle returns 0. If $b = 1$, $\text{Dec}_{\text{sk}}(c_1) = B$. In that case, the oracle returns a message $m \neq 0$. This allows us to distinguish the cases and win the IND-CPA^D game. $\square$

C Technical Lemmata

In this section, we generalize Lemma 3.6 and 3.7 from [8] to the matrix setting.

Lemma C.1 ([26], Corollary 2.8). *Let $\Lambda \subseteq \mathbb{Z}^m$ be a lattice, $0 < \varepsilon < 1$, and $r > 0$. For any vector $\mathbf{c} \in \mathbb{R}^m$, if $r \geq \eta_\varepsilon(\Lambda)$, then we have*

$$\rho_r(\Lambda + \mathbf{c}) \in \left[\frac{1 - \varepsilon}{1 + \varepsilon}, 1 \right] \cdot \rho_r(\Lambda).$$

Lemma C.2 ([4], Lemma 2.13). *For any rank- n lattice $\Lambda \subset \mathbb{R}^n$ and $\varepsilon \in (0, 1/2)$,*

$$\lambda_n(\Lambda) \leq \sqrt{n} \cdot \eta_\varepsilon(\Lambda).$$

For any $\mathbf{A} \in \mathbb{Z}_q^{n \times k}$, $\mathbf{c} \in \mathbb{Z}^k$, let $\hat{\Lambda} = \Lambda_q^\perp(\mathbf{A}) \times \mathbb{Z}^m$ and $\hat{\mathbf{c}} = (\mathbf{c}, 0) \in \mathbb{Z}^{k+m}$. For $\mathbf{E} \in \mathbb{Z}^{m \times k}$ a full row-rank matrix, let $\hat{\mathbf{E}} = [\mathbf{E} | \mathbf{I}_m] \in \mathbb{Z}^{m \times (k+m)}$ and $r > 0$, we consider

$$\mathcal{E}_{\mathbf{E}, r, \mathbf{c}} := \{\hat{\mathbf{E}}\mathbf{v} \mid \mathbf{v} \leftarrow \mathcal{D}_{\hat{\Lambda} + \hat{\mathbf{c}}, r}\} \quad \mathcal{L}_{\mathbf{A}, \mathbf{E}} := \{\mathbf{v} \in \hat{\Lambda} : \hat{\mathbf{E}}\mathbf{v} = 0\}.$$

Lemma C.3 (adapted from [8], Lemma 3.6). *Let $\varepsilon, r > 0$. For any $\mathbf{A} \in \mathbb{Z}_q^{n \times k}$, $\mathbf{c} \in \mathbb{Z}^k$ and $\mathbf{E} \in \mathbb{Z}^{m \times k}$ a full row-rank matrix, let $\mathcal{L}_{\mathbf{A}, \mathbf{E}}$ be as defined above. If $r \geq \eta_\varepsilon(\mathcal{L}_{\mathbf{A}, \mathbf{E}})$, then*

$$\Delta(\mathbf{E}\mathbf{x} + \mathbf{y}, \mathbf{e}') < 2\varepsilon$$

where $\mathbf{x} \leftarrow \mathcal{D}_{\Lambda_q^\perp(\mathbf{A}) + \mathbf{c}, r}$, $\mathbf{y} \leftarrow \mathcal{D}_{\mathbb{Z}^m, r}$, and $\mathbf{e}' \leftarrow \mathcal{D}_{\mathbb{Z}^m, r\hat{\mathbf{E}}^\top}$.

The arguments from [8, Lemma 3.6] get adapted to the matrix setting using standard techniques by considering the pseudoinverse of $\mathbf{E}^\top \mathbf{E}$ instead of the inverse of $\|\mathbf{e}\|^2$. The proof combines the techniques from [8] to show the independence from c , with techniques from [2,1] for the matrix results.

Proof. Let $\hat{\mathbf{E}} = [\mathbf{E} | \mathbf{I}_m] \in \mathbb{Z}^{m \times (k+m)}$, $\hat{\mathbf{c}} = (\mathbf{c}, 0) \in \mathbb{Z}^{k+m}$ and $\hat{\Lambda} = \Lambda_q^\perp(\mathbf{A}) \times \mathbb{Z}^m$. Consider the distribution

$$\mathcal{E}_{\mathbf{E}, r, \mathbf{c}} = \{\hat{\mathbf{E}}\mathbf{v} \mid \mathbf{v} \leftarrow \mathcal{D}_{\hat{\Lambda} + \hat{\mathbf{c}}, r}\}.$$

Fix some $\mathbf{z} \in \mathbb{Z}^m$, and let $\mathcal{L}_{\mathbf{z}} := \{\mathbf{v} \in \hat{\Lambda} + \hat{\mathbf{c}} : \hat{\mathbf{E}}\mathbf{v} = \mathbf{z}\}$. The probability mass assigned to $\mathbf{z}$ by $\mathcal{E}$ is the mass assigned to the coset $\mathcal{L}_{\mathbf{z}}$ by $\mathcal{D}_{\hat{\Lambda} + \hat{\mathbf{c}}, r}$. Therefore it is proportional to $\rho_r(\mathcal{L}_{\mathbf{z}})$.

The lattice $\mathcal{L}_{\mathbf{A}, \mathbf{E}} = \{\mathbf{v} \in \hat{\Lambda} : \hat{\mathbf{E}}\mathbf{v} = 0\}$ is orthogonal to the rows of $\hat{\mathbf{E}}$. Notice that for any $\mathbf{w}_{\mathbf{z}} \in \mathcal{L}_{\mathbf{z}}$, we have that $\mathcal{L}_{\mathbf{z}} = \mathcal{L} + \mathbf{w}_{\mathbf{z}}$. The support of the distribution $\mathcal{E}$ is $\mathbf{E}\hat{\Lambda} + \mathbf{E}\hat{\mathbf{c}} + \mathbb{Z}^m = \mathbb{Z}^m$, so $\mathbf{w}_{\mathbf{z}}$ exists.

Let $\mathbf{J}$ be the pseudoinverse of $\hat{\mathbf{E}}$. Let $\mathbf{t} \in \mathcal{L}_{\mathbf{z}}$. Let $\mathbf{u}_{\mathbf{z}} = \mathbf{J}^\top \mathbf{z}$. Note that $\mathbf{u}_{\mathbf{z}}$ is orthogonal to $\mathcal{L}$. In particular, $\mathbf{u}_{\mathbf{z}}$ is orthogonal to $\mathbf{t} - \mathbf{w}_{\mathbf{z}}$. It follows that:

$$\rho(r^{-1}\mathcal{L}_{\mathbf{z}}) = \rho(\mathbf{u}_{\mathbf{z}}) \cdot \rho(r^{-1}\mathcal{L}_{\mathbf{z}} - \mathbf{u}_{\mathbf{z}}).$$

The rest of the proof follows as in [8]. For $r \geq \eta_\varepsilon(\mathcal{L}_{\mathbf{A}, \mathbf{E}})$, it follows from Lemma C.1 that,

$$\rho(r^{-1}\mathcal{L}_{\mathbf{z}}) = \left[\frac{1-\varepsilon}{1+\varepsilon}, 1 \right] \cdot \rho_r(\mathcal{L}) \cdot \rho(\mathbf{z}).$$

Hence the statistical distance between $\mathcal{D}_{\mathbb{Z}^m, r\hat{\mathbf{E}}^\top}$ and $\mathcal{E} = \hat{\mathbf{E}}\mathcal{D}_{\hat{\Lambda} + \hat{\mathbf{c}}, r}$ is at most $1 - \frac{1-\varepsilon}{1+\varepsilon} \leq 2\varepsilon$. $\square$

If we set $\mathbf{A} = \mathbf{G}$, we obtain the following corollary.

Corollary C.1. *Let $\varepsilon, r > 0$. Let $\mathbf{E} \in \mathbb{Z}^{m \times k}$ be a full row-rank matrix and $\mathbf{c} \in \mathbb{Z}^m$, if $r \geq \sqrt{5}(1 + s_1(\mathbf{E}))\sqrt{\frac{\ln 2k(1+1/\varepsilon)}{\pi}}$, then*

$$\Delta(\mathbf{E}\mathbf{x} + \mathbf{y}, \mathbf{e}') < 2\varepsilon$$

where $\mathbf{x} \leftarrow \mathcal{D}_{\Lambda_q^\perp(\mathbf{G})+\mathbf{c}, r}$, $\mathbf{y} \leftarrow \mathcal{D}_{\mathbb{Z}^m, r}$, and $\mathbf{e}' \leftarrow \mathcal{D}_{\mathbb{Z}^m, r\hat{\mathbf{E}}^\top}$.

Proof. This follows directly from Lemma C.3 and Lemma C.5.

The following two lemmata bound the smoothing parameters of the $\mathcal{L}_{\mathbf{A}, \mathbf{E}}$ and the $\mathcal{L}_{\mathbf{G}, \mathbf{E}}$ lattices.

Lemma C.4. *Let $\varepsilon > 0$. Let $k \geq n \log q + \omega(\sqrt{\log n})$. For $\mathbf{E} \in \mathbb{Z}^{m \times k}$ a full row-rank matrix and $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_q^{n \times k}$, let $\mathcal{L}_{\mathbf{A}, \mathbf{E}}$ be as defined above. We have*

$$\eta_\varepsilon(\mathcal{L}_{\mathbf{A}, \mathbf{E}}) \leq k \cdot 2\eta_\varepsilon(\mathbb{Z})(1 + s_1(\mathbf{E}))\sqrt{\frac{\ln 2k(1 + 1/\varepsilon)}{\pi}}$$

We essentially follow the proof of [8, Lemma 3.7], replacing the explicit basis for $\Lambda_q^\perp(\mathbf{G})$ by a generic bound.

Proof. With overwhelming probability, $\Lambda_q^\perp(\mathbf{A})$ is a rank- k lattice. We have that $\lambda_k(\Lambda_q^\perp(\mathbf{A})) \leq \sqrt{k} \cdot \eta_\varepsilon(\Lambda_q^\perp(\mathbf{A})) \leq 2\eta_\varepsilon(\mathbb{Z})$ where the first inequality follows from Lemma C.2 and the second from Lemma 2.2. It follows that there exists a basis $\bar{\mathbf{B}}$ for $\Lambda_q^\perp(\mathbf{A})$ such that,

$$\begin{aligned} \|\bar{\mathbf{B}}\| &\leq \sqrt{k}\lambda_k(\Lambda_q^\perp(\mathbf{A})) \\ &\leq k \cdot 2\eta_\varepsilon(\mathbb{Z}) \end{aligned}$$

The lattice $\mathcal{L}_{\mathbf{A}, \mathbf{E}} = \hat{\Lambda} \cap \ker \hat{\mathbf{E}}$, so $\mathcal{L}_{\mathbf{A}, \mathbf{E}}$ has rank k . The rest of the proof is unchanged from [8, Lemma 3.7] and the lemma follows.

Lemma C.5. *Let $\varepsilon > 0$. For $\mathbf{E} \in \mathbb{Z}^{m \times k}$ a full row-rank matrix, let $\mathcal{L}_{\mathbf{G}, \mathbf{E}}$ be as defined above. We have*

$$\eta_\varepsilon(\mathcal{L}_{\mathbf{G}, \mathbf{E}}) \leq \sqrt{5}(1 + s_1(\mathbf{E}))\sqrt{\frac{\ln 2k(1 + 1/\varepsilon)}{\pi}}$$

The proof follows directly from the proof of [8, Lemma 3.7], simply replacing $\hat{\mathbf{e}}$ with $\hat{\mathbf{E}}$ when defining the shortest vectors $\mathbf{u}_k$.